Resource Extraction and Conversion from Lunar Anthropogenic Inputs with Microwaves

The Pennsylvania State University

Faculty Advisors:

Dr. Sven Bilén	The Pennsylvania State University
Dr. Aleksandra Radlińska	The Pennsylvania State University

RECLAIM Team Leaders:

Charles McManus	Aerospace Engineering
Julian Ruocco	Aerospace Engineering

RECLAIM Team Members:

Aum Dave	Aerospace Engineering
Noah Ferris	Electrical Engineering
Joseph Flanagan	Aerospace Engineering
Ammar Hattab	Aerospace Engineering
Sam Mauro	Aerospace Engineering
Yash Miglani	Aerospace Engineering
Joe Parker	Aerospace Engineering
Olivia Schaffer	Aerospace Engineering
Joseph Schmidt	Engineering Science and Mechanics
Marisa Tommarello	Aerospace Engineering
Luke Waszyn	Engineering Science and Mechanics
Jackson Wetzel	Aerospace Engineering
Lucas Wynman	Electrical Engineering
Abigail Ziegenfuss	Aerospace Engineering

Contents

1. Vision and Innovation **4**

1.1. Overview of the Solution 4

1.2. Innovation and Challenge Impact 5

1.3. Improvement over Current Technologies 7

1.4. Advanced Technologies Used 8

1.5. Commercial Potential 9

1.6. How could your design be adapted for application in a future lunar mission and/or on the lunar surface 11

2. Final Round Solution Design **12**

2.1. Final Round Engineering Design 13

2.1.1. Design Rationale 13

2.1.2. Performance Analysis 20

2.1.3. System and Component Specifications 25

2.2. Final Round Operations, Analyses, and Test Results 27

2.2.1. Concept of Operations 27

2.2.2. Inputs, Outputs, and Efficiency 31

2.2.3. Output Characterization and Verification 32

2.3. Final Round Schematics 34

2.4. Final Round Master Equipment List 42

3. Pre-Processing of Trash **45**

3.1. Demonstrated Pre-Processing 45

3.2. Pre-Processing Inputs Table 46

FINAL ROUND JUDGING CRITERIA

Criteria	Description	Points
Completeness of the Submission	Did the team address all of the requirements listed in the submission template?	Pass/Fail
Vision and Innovation	• How well has the team articulated the vision and innovation for the solution? • How does the solution build or improve upon the current state of the art? • How does it leverage advanced technologies? • What is the commercial potential of the solution (terrestrial and/or in space)?	Up to 15 points
Feasibility	• How feasible will the design be to build and implement for a future NASA mission? • How well does the team understand and articulate potential technical risks and how they could be mitigated?	Up to 15 points
System Operating Requirements	• How well has the team minimized the mass and volume of the system? • What is the Energy Efficiency of the system? • What is the Mass Efficiency of the system? • How well has the team minimized the crew time necessary to operate and maintain the solution? • How well has the team minimized the production of hazardous byproducts, including particulate matter such as microplastics or metal shreds?	Up to 15 points
Net Outputs	• How useful are the outputs from the recycling process (feedstocks and/or end products)? • How well has the team described their method of characterizing and ensuring the quality of outputs?	Up to 5 points
Performance in the Demonstration	• How many types of eligible trash items (from one to eight, see Table 4) did the team recycle? • What is the Energy Efficiency of the system? • What is the Mass Efficiency of the system? • How many team members were required? How many minutes of time per team member were required? • Did the usable outputs meet characterization or quality standards, as defined by the teams?	Up to 50 points
	TOTAL	**Up to 100 points**

1. Vision and Innovation

JUDGING CRITERIA

- How well has the team articulated the vision and innovation for the solution?
- How does the solution build or improve upon the current state of the art?
- How does it leverage advanced technologies?
- What is the commercial potential of the solution (terrestrial and/or in space)?

WRITING REQUIREMENTS

1. 1.1. Provide a brief overview of your solution, including eligible trash items that you will recycle (from TABLE 4); overview of the recycling system; and outputs (both usable and unusable).
2. 1.2. How does your solution use innovation to address the challenge?
3. 1.3. How does the solution build or improve upon the current state of the art?
4. 1.4. How does your solution leverage advanced technologies?
5. 1.5. Describe the commercial potential of your solution.
6. 1.6. How could your design be adapted for application in a future lunar mission and/or on the lunar surface?

1.1. Overview of the Solution

Req: Provide a brief overview of your solution, including eligible trash items that you will recycle (from TABLE 4); overview of the recycling system; and outputs (both usable and unusable).

- Explanation of RECLAIM system
  - Briefly list the materials that can be recycled
- Integration of RECLAIM into ILUMEN
  - Save the "full explanation of ILUMEN" until 1.2 — briefly state it is part of a larger system
- Explain scope of products that are being produced and how they are useful
  - Explain usefulness of carbon char, oil, and syngas — make sure to mention how RECLAIM produces them but does not directly refine them
  - Introduce subsystems and their innovations before introducing them **[charles 7/20]** this doesn't need to happen here, but it should happen

The Resource Extraction and Conversion from Lunar Anthropogenic Inputs with Microwaves (RECLAIM) system is an integrated application system of planned lunar infrastructure that will provide means of recycling waste stream materials from a future lunar colony.

In the context of the LunaRecycle Challenge, this system is known as RECLAIM*Lite*. RECLAIM*Lite* represents terrestrial limitations to the system, but serves to demonstrate what a future lunar integration could look like as demonstrated in Figure 1.

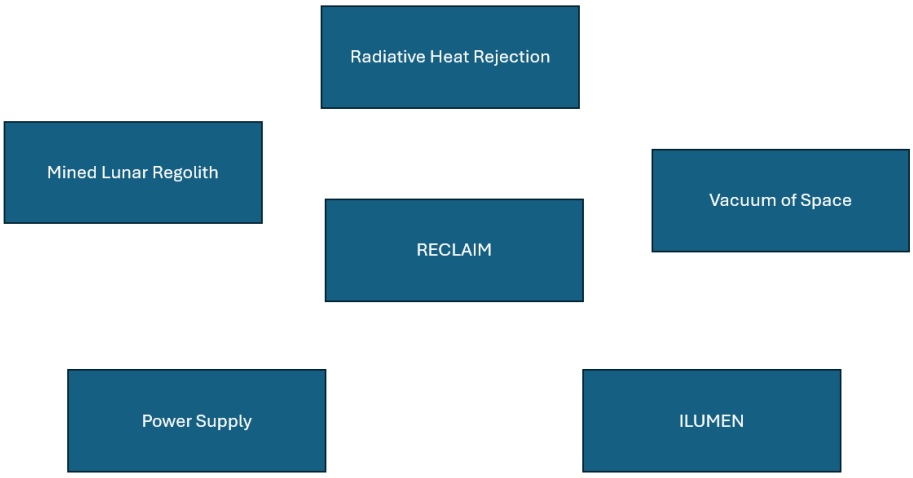


Figure 1: RECLAIM System preliminary idea

[fortnite 7/21] what is this figure trying to show, is it saying that all of those things are outside the scope of reclaim? maybe have arrows pointing into reclaim

The eligible waste items that RECLAIM*Lite* recycles during the LunaRecycle competition, along with the subsystem that processes each and its resulting outputs, are summarized in the table below.

RECLAIM Recycling Overview

Eligible Trash Item (Table 4)	**Material Composition**	**RECLAIM Subsystem**	**Usable Outputs**	**Unusable Outputs**
Aluminum structure/struts	Aluminum >90% (6061)	SMELT (microwave smelting & drip casting)	Cast aluminum parts and ingots/billets	Residual slag retained in crucible
Reclosable bag	Polyethylene 100%	Plastics pyrolysis	Syngas, pyrolysis oil, wax, and carbon char	Waxy residue retained in lines; spent purge gas

1.2. Innovation and Challenge Impact

Req: How does your solution use innovation to address the challenge?

- One closed loop process for two different waste stream materials
- ILUMEN + closed loop water supply + utilizing lunar environment

RECLAIM is a innovative system supporting NASA's goal of establishing a long-term self-sustaining human presence on the Moon via its Artemis program. RECLAIM is an application system of the ILUMEN architecture, which stands for Infrastructure for Lunar Utilization of Microwave Energy. ILUMEN utilizes a high-powered energy source to produce microwave energy, which can then be used for a variety of applications like RECLAIM as seen in Figure 2.

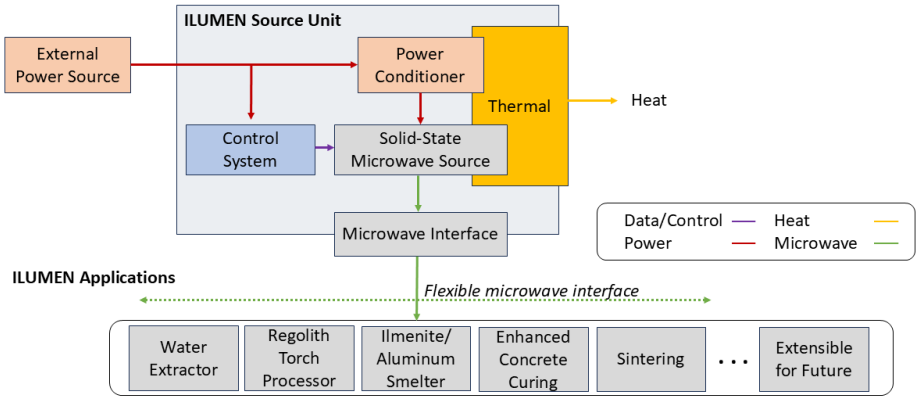


Figure 2: ILUMEN architecture Bilén, 2024.

To expand and become more independent, a lunar colony must develop robust industrial processes for habitat construction, life support (water and oxygen), and agriculture. The ILUMEN system will help power this infrastructure by leveraging microwaves, which are highly efficient in this environment due to negligible transmission loss within the lunar exosphere. Complementing this, the RECLAIM system focuses on in-situ resource utilization (ISRU) to manufacture metallic components and supplement fuel production, drastically reducing Earth-to-Moon launch costs.

RECLAIM ties into this lunar industrial complex by recycling key waste items that colonists will use and will have to dispose of. Plastics such as polyethylene from food packaging and aluminum from one-way spacecraft will be among the waste items generated by future colonists and explorers. RECLAIM is designed to accept a wide variety of feedstock that can increase the inherent value of otherwise disposable items brought to the lunar surface, as well enable a closed-loop economy by mitigating the need for lunar landfills. The combined ability of metal smelting and pyrolysis gives RECLAIM a large material operation that extends past the waste items described within the LunaRecycle challenge. SMELT Team, 2023.

RECLAIM continues to innovate by building off of existing NASA architecture by structuring itself within the BliSS rack. The BliSS rack is a modified version of the EXPRESS rack that is used on the ISS, utilizing the same utility-rack design but with a rectangular form factor that could be easily shipped and wheeled into a lunar habitat (Walter et al., 2025)

RECLAIM also utilizes ISRU technology and research by implementing in-situ materials. For example, Ilmenite is utilized as both a catalyst for pyrolysis of plastic and as a sand-like substitute for sand-casting.

[julian 7/21] Too Long move to design rationale?, Ilmenite was selected due to its presence on the lunar surface, its explored use in existing water-reclamation technologies, and its iron content. Ilmenite can act as a catalyst for cleaving long carbon chain molecules that are present in many waste stream materials by

reducing the activation energy required to break the C-C and C-H bonds. Its usefulness in lunar industrial processes is thus being further explored through RECLAIM. **[fortnite 7/21]** i think it is important to highlight that it is already being used for other stuff, so it wouldn't require an entire additional system

1.3. Improvement over Current Technologies

Req: How does your solution leverage advanced technologies?

- SSMG compared to magnetrons
  - Transistors instead of vacuum tube
- Compare microwave heating to other conventional heating methods
- Comparing drip cast to manual pour
- SSMG
  - Solid state transistors (GaN)
  - Heating behavior/physics (volumetric and selective)
  - Precise control over power and frequency
- Pyrolysis
  - Microwave-assisted pyrolysis (frequency variation)
  - Ilmenite catalysis + ZSM-5 / synthetic catalysis as future work

RECLAIM uses advanced, state-of-the-art gallium nitride (GaN) based solid-state microwave technology to efficiently pyrolyze plastic and smelt aluminum. At its core, it leverages ILUMEN's electrically powered SSMG that converts electricity into high-power radio frequency (RF) energy in the microwave band (2.4–2.5 GHz) with a maximum output power of 6.0 kW.

SSMG technology is unlike conventional magnetron based generators that operate within a relatively narrow frequency range and can drift due to changes in the load that could result in the formation of hotspots in a process chamber. The SSMG provides precise control of frequency, phase, and output power which improves heating uniformity (Zhou et al., 2023). Solid-state sources have several other advantages over conventional magnetron microwave generators: low-voltage operation, frequency flexibility, and much longer lifetimes. Furthermore, the modular design of many solid-state sources also enables easier troubleshooting and repairs compared to magnetrons which have sensitive parts such as heated filaments (RFHIC Corporation, 2021).

The SSMG is able to convert DC power into high frequency microwave power using transistors. Recent advances in solid-state microwave technology have enabled the use of GaN transistors. These advances have improved drain efficiencies and power densities, allowing solid-state sources to become more compact and offer higher power outputs than previously (Yamanaka et al., 2024). Improvements with GaN transistors are closing the gap between the typically higher efficiencies seen in magnetrons when compared to solid state sources. These advances indicate that solid state sources are increasingly more competitive with magnetrons while maintaining advantages in power and frequency control and operational lifetime. **[Noah 7/21]** This last sentence is very similar to the previous paragraph's last sentence, couple the last two paragraphs into one, decrease overall size, convey ILUMEN as the centralized source as an SSMG?

For comparison, conventional furnace-based heating methods (e.g., electric and hydrocarbon-based) require high power levels and typically require substantial supporting infrastructure to generate, contain, and manage heat. Depending on the heating process, this may include material charging and handling equipment, off-gas collection and treatment, or replacement of worn furnace components. Such additional requirements make the operation of conventional systems difficult in a lunar environment. In contrast, RECLAIM*Lite* recycles metals using electrical power and a closed loop cooling system at lower energy consumption. In addition, microwaves offer more uniform and efficient heating of the load through selective and volumetric heating (Effinger et al., 2022). This not only reduces the time required for operation, but also increases safety.

RECLAIM utilizes a modified continuous casting method implementing a bottom-pour crucible within the SMELT subsystem. Heritage systems and other forges for small-scale smelting requires the user to use extensive personal protective equipment (PPE) and manually retrieve a crucible from a high-temperature chamber to cast it. This process poses substantial safety and crew-time concerns that the continuous casting adaptation aims to mitigate. RECLAIM's adapted continuous casting eliminates direct human interaction with high temperature equipment and molten metal thus minimizing apparent safety concerns.

1.4. Advanced Technologies Used

Req: How does your solution leverage advanced technologies?

- SSMG
  - Solid state transistors
  - Heating behavior/physics
  - Precise control over power and frequency
- Pyrolysis
  - Microwave-assisted pyrolysis (frequency variation)
  - Ilmenite catalysis **[charles 7/15]** I am putting ilmenite catalysis in 1.2 instead of 1.4 because 1.2 is going in-depth with the whole ISRU story **[julian 7/19]** Ok, this is cool. I think you could introduce in 1.2 both BLiSS and Catalysts and delve deeper into them for 1.6.

RECLAIM*Lite* uses a solid-state microwave generator (SSMG) that ILUMEN would provide for RECLAIM to enable efficient in-situ material processing and recycling on the lunar surface. The SSMG provides precise control over the microwave energy parameters that allow for corrections during the process.

Microwave heating was selected because it provides several advantages over alternative heating solutions that could be used on the lunar surface. Microwaves heat material volumetrically, allowing energy to be deposited into a microwave absorbent region rather than relying on heat transfer from the processing chamber's walls seen in surface heating methods (Zhou et al., 2023). Conventional pyrolysis typically operates at heating rates of approximately 5 to 20 °C/min, whereas microwave assisted pyrolysis can achieve significantly higher heating rates of approximately 50 to 100 °C/min (Bai et al., 2026). This reduces the time needed to heat up a chamber and reduces thermal losses, increasing overall energy efficiency.**[charles 7/20]** explain why volumetric heating is advantageous over surface heating **[wetz 7/21]** Entire last section dedicated to explaining advantages over current . This is problem I run into because one way or another I'm going to be repeating myself to an extent fortnite- I think this paragrpah is good enough, wouldn't need to be explained

again later

Materials with lossy dielectric properties absorb electromagnetic energy and convert it into heat, whereas highly conductive materials tend to reflect most of the incident energy. This selective heating behavior is beneficial during aluminum smelting when energy from the microwaves is absorbed directly by the silicon carbide (SiC) crucible. This is optimal for the smelting process, as the crucible acts as a microwave absorbing susceptor, efficiently transferring heat to the aluminum. This can reduce the required heating time and the thermal losses associated with heating the entire chamber structure (Jiang et al., 2024). **[wetz 7/20]** Thinking 2nd and 3rd paragraphs could get combined (microwave heating behavior/physics) **[charles 7/20]** I think again this is kind of a generic statement and does not fully explain why volumetric microwave heating is beneficial fortnite- i think it is fine now with the citation

An SSMG is particularly advantageous for pyrolysis because it provides precise control over various microwave parameters, including adjustable output power and an operating frequency within the range of 2.40 to 2.50 GHz, which straddles the same frequency band used by household microwave ovens (2.45 GHz). When utilizing a single fixed frequency, standing wave patterns can form, thus creating localized hot and cold spots in the chamber (Bai et al., 2026). By varying or sweeping the frequency of the microwave energy, the electromagnetic field patterns shift during operation, effectively evenly distributing the deposited energy. In addition, as the processing load pyrolyzes, its microwave absorption behavior and dielectric properties may change as well. The ability to alter frequency can improve microwave coupling and promote more uniform heating as the material's properties change, increases the overall efficiency of the process. **[fortnite 7/21]** important to the design, but not important to the results or outcome. not mentioning this doesn't really change the understanding of the system. Maybe remove or shorten to make room. change next paragraph based on the decision here

This unique heating solution decreases overall system complexity by reducing the amount of physical hardware, instead replacing it with state-of-the-art microwave component to achieve greater effect.

1.5. Commercial Potential

Req: Describe the commercial potential of your solution.

- Argue why it is useful for terrestrial scenarios
- Argue for lunar scenarios
- Ability to make any desired part

RECLAIM situates itself well for commercial viability by housing a fully integrated, modular, portable recycling system that can be used in lunar applications. The implementation of the RECLAIM system would also provide valuable knowledge to the recycling industry based off the performance of microwave-based recycling, which is a highly active area of research **[fortnite 7/19]** this is a good place to put any microwave related pyrolysis papers that were used/referenced.

Plastic pyrolysis is not commercially common on Earth due to the over-abundance of ethylene making it cost, which is a product of ethane. Ethane is the sister product of methane from the natural gas industry, which is considered "waste product" itself. The over-supply of ethane makes the production of new plastic extremely cheap. Recycling plastic by decomposition is an energy intensive process and thus is not currently economical on Earth. **[charles 7/19]** I am pretty sure this is correct, but will need to double check

However, the Moon presents a unique need for and opportunity to utilize plastic pyrolysis to generate a wide-range of products. The production of hydrocarbons (particularly methane, needed as rocket fuel for methalox engines), oil, carbon char, hydrogen, and other syngas outputs would be challenging to make via other ISRU processes on the Moon, but pyrolysis at small scales allows a lunar colony to supplement itself with these products that would otherwise need to be shipped from Earth.

As an example, carbon char, which is generated by the pyrolysis system through the degradation of carbonaceous waste, can be turned into activated carbon, which is a highly versatile resource in the lunar environment, particularly for air and water filtration. Similarly, pyrolyzing biological matter such as inedible biomass or fecal matter, can produce biochar which can supplement agriculture development. **[Noah 7/21]** Can this paragraph be combined with the last, as they both give examples of what can be used via plastic pyrolysis McCoy, 2012

Additional pyrolysis products such as hydrogen, methane, lubricating oil, and paraffin wax all have separate applications with important use cases. The absence of these resources in the lunar environment creates a significant need for methods with which they can be made on the Moon.

With RECLAIM being a system designed to recycle not just plastics, but metals as well, materials that seemed to be one-time use before can be repurposed and recycled. The versatility of a sand mold offers the option of creating any number of tools or pieces needed on the lunar surface. Hand held tools like wrenches and screwdrivers can be produced in substitution of a broken tool. A tool breaking could mean the loss of operations for an astronaut. RECLAIM eliminates that holding time to receive a new tool. In addition to tools, RECLAIM can cast aluminum ingots that align with current technology used at NASA Marshall. There, they employ an aluminum die extrusion system called LEIA by Cislunar Industries. RECLAIM can supply the billets used by this system, making it compatible with NASA's existing in-space manufacturing approach. **[fortnite 7/19]** aluminum or multiple metals? **[olivia 7/21]** @fortnite I said metals generally because we said we can process ilmenite**[olivia7/20]** STILL WORKING ON THIS SECTION **[fortnite 7/19]** we are only using aluminum so maybe we shouldn't focus too much on tools **[olivia 7/21]** i think we can keep the tools part because they still work, just aren't as strong as typical tools ofc. can be useful if something breaks and they need a quick replacement part. i think the LEIA/ billets is really appealing as it appeals to a technology they are already using. I can talk to Bilen more and go more in depth about what LEIA is if necessary.

[charles 7/19] also need to mention terrestrial impacts like the Phase I report for military bases, remote communities, that kind of thing

[julian 7/19] this section needs signifcant work, arguing commercial potential on earth requires a more fleshed out response. Pyrolysis itself is inherently not financially feasible in the current market. Selling the commercial potential should revolve more around ILUMEN, launch costs, and remote military installations. I thought that this section in the previous report was much stronger.

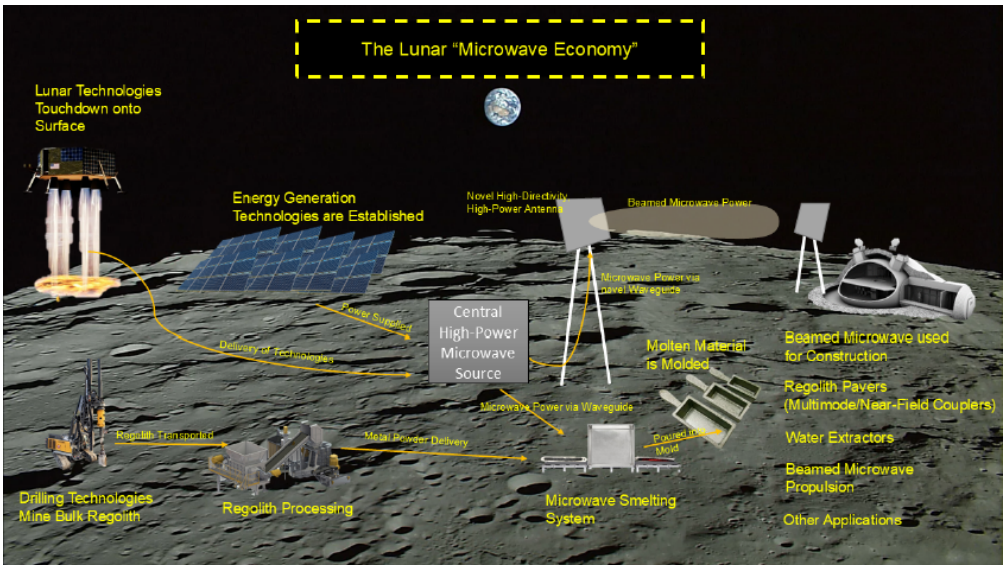


Figure 3: lunar economy figure Bilén, 2024

1.6. How could your design be adapted for application in a future lunar mission and/or on the lunar surface

- How RECLAIM attempted to implement ISRU materials
- RECLAIM aiming for Phase II / Phase III implementation compared to most competitors geared for short term Phase I
- SiC ISRU for both chambers using Sabatier methane
- Methane for microbial biomanufacturing
- Ilmenite casting
- Ilmenite for catalysts
- Distillation for separating compounds?
- Elimination of oxidation
- Combustion prevention
- Use of vacuum
- ILUMEN
- BLISS rack implementation
- **[charles 7/19]** Lets assess if we should include "the system could be optimized to exist on the lunar surface as future work" and include some benefits. It might be advantageous, but is also not in the context of LunaRecycle challenge
- Explain use of vacuum for vacuum-assisted pyrolysis

NASA defines 3 key phases in reaching continuous human presence on the Moon. NASA, 2026 RECLAIM aligns best with Phases 2 and 3, providing NASA with a long-term recycling system that supports human presence and development on the Moon.

RECLAIM is designed to be operated in the lunar environment. The current system functions within the challenge-designated ambient conditions found on Earth, but iterating the design would involve removing the terrestrial limitations of the system. The vacuum pump and chiller would be removed to instead utilize facilities already present on the lunar surface such as the vacuum of the Moon and radiative cooling infrastructure. RECLAIM exists as an application system of ILUMEN, further serving lunar integrated infrastructure goals.

Ilmenite is used for two purposes in RECLAIM and, as such, is an example of ISRU within the system. Ilmenite is a dense iron–titanium oxide and is the fourth most abundant mineral found in lunar regolith. RECLAIM utilizes this mineral as a catalyst within its plastic pyrolysis system and as a sand-casting medium.

RECLAIM utilizes an ilmenite-based mixture as a substitute for traditional casting sand. When ilmenite is combined with 40% sodium silicate and water at a 1:2 ratio, it creates moist sand similar to that of conventional green casting sand. Following the casting process, the ilmenite mold is broken apart and sifted to recover the fine material, which can then be reactivated and reused in subsequent castings. Sodium silicate is used as the binder because it also serves as an alkali activator in the production of regolith-based lunar concrete, making it a readily available resource within the lunar infrastructure (Zheng et al., 2024).**[charles 7/20]** add ref

Silicon carbide (SiC) is an additional material used in RECLAIM that demonstrates preliminary ISRU compatibility. Production of SiC for use in microwave engineering using carbothermal reduction is an active area of research (Srimurugan and Subbiah, 2025). **[charles 7/20]** going to have to change this around wording-wise

RECLAIM is currently proposed to be located inside of a lunar habitat, although further development could enable placement external to habitats. **[fortnite]** this was already said 4 paragrpahs ago. combine the two? In the space environment, the elimination of oxygen could improve the quality of metal casting due to the lack of oxidation, and eliminate the need of a purge gas to prevent combustion during pyrolysis. **[charles 7/20]** this is a giant can of worms that could be opened, I think we should just keep it brief like this for now, and maybe include some caveats to this consideration

[?] Can Mention - Microwave Energy Transfer - How an outside vacuum could be useful for pyrolysis as well. - Cooling loop underground - Im sure there are many more Pro's (and cons)

2. Final Round Solution Design

JUDGING CRITERIA
Feasibility

- How feasible will the design be to build and implement for a future NASA mission?
- How well does the team understand and articulate potential technical risks and how they could be mitigated?

System operating requirements

- How well has the team minimized the mass and volume of the system?

- What is the Energy Efficiency of the system?
- What is the Mass Efficiency of the system?
- How well has the team minimized the crew time necessary to operate and maintain the solution?
- How well has the team minimized the production of hazardous byproducts, including particulate matter such as microplastics or metal shreds?

Net Outputs

- How useful are the outputs from the recycling process (feedstocks and/or end products)?
- How well has the team described their method of characterizing and ensuring the quality of outputs?

WRITING REQUIREMENTS — 2.1. Final Round Engineering Design (Recommended length: 6–8 pages)

1. 2.1.1. A rationale for your design approach
2. 2.1.2. Preliminary evidence and analysis predicting performance including mass, volume, and power requirements
3. 2.1.3. System-level and component-level design specifications for hardware and software, including bill of materials and system mechanical schematic. This section should also include the maximum allowable temperature (°C) and maximum allowable working pressure (PSIG) for the equipment.

2.1. Final Round Engineering Design

Recommended length for 2.1: 6–8 pages

2.1.1. Design Rationale

Req: A rationale for your design approach.

2.1.1.1. Microwave Subsystem Design

- SSMG to provide energy
- Circulator & isolator protect from RP
- Tuner to maximize FP
- Switch to allow 2 processes with 1 SSMG
- Cooling loop to allow for high-powered nature (cooling jacket)
- Argue in a perfect world, no adapters and less waveguide pieces needed with an unlimited budget

The RECLAIM system aims to fit within the scope of ILUMEN's centralized microwave source. RECLAIM*Lite* aims to demonstrate just one of these potential use cases for the ILUMEN concept which fits within the LunaRecycle challenge. In the microwave subsystem, microwave power is generated from the SSMG which passes through a circulator, automatic tuner, and microwave switch before being routed to the selected chamber. A dummy load is also connected to the third port of the circulator to absorb the reflected waves, protecting the SSMG.

Microwave heating was selected for this system due to the unique behavior that allows for efficient and uniform heating. The SSMG in particular offers both selective behavior and volumetric heating while also allowing precise control over microwave power and frequency. The operator can gradually increase power, adjust the heating rate, and turn off microwave power if there are unsafe operating conditions.

[wetz 7/20] Bilen said important to address that the parts used for the challenge aren't neccessarily the parts used for the RECLAIM system that would be sent to the Moon. Address that certain limitations with components are challenge specific. May only need to address the actual component specification in BOM and schematics and explain what those components do in a general way where certain limiting factors we have in the challenge aren't in the scope of RECLAIM's vision as a wholeThe Daihen SMA-15B Automatic Tuner is a component of the RECLAIM*Lite* system that adjusts the impedance match between the SSMG and the selected load. This specific tuner is powered through a Daihen CMA-10 controller which provides power and controls to the tuner's motors to move three metal stubs that adjust impedance between the SSMG and load. Microwave power absorption is strongly influenced by the number of tuning stubs, increasing from approximately 3% without a stub to 37% with one stub and up to 80% with four stubs (Ellison et al., 2026). **[wetz 7/21]** May not be important enough to include but wanted to explain rationale for using multi-stubbed tunerDuring operation, the impedance of the load can change as the material is melted or pyrolyzed. The automatic tuner responds by adjusting the impedance match between the load and the microwave transmission system, increasing power transfer and reducing reflected power. (Liu et al., 2024).

The RECLAIM*Lite* system utilizes a WR430 Muegge-Gerling GA1232 Water Load connected to a WR340 Astex Ferrite Circulator via a waveguide adapter to protect the SSMG from reflected microwave power. Despite utilizing an automatic tuner to reduce most of the reflected microwave power, some power may reflect back toward the SSMG during operation. The circulator allows forward microwave power to continue on its path toward the tuner. Using magnetically biased ferrite material, the current isolator then redirects reflected microwave power to the dummy load. **[7/21 wetz]** RECLAIM*Lite* uses ferrite circulator but is this neccessary to include if RECLAIM won't neccessarily be using a ferrite circulatorThe dummy load then safely absorbs this power as heat, improving the reliability and safety of the microwave subsystem (Gou et al., 2023).

The WR284 Logus MM284APLx59 switch is included in the RECLAIM*Lite* system to route power to the selected chamber, allowing the microwave source to initiate the pyrolysis or smelt process. A switch will be included in the RECLAIM system to reduce complexity in comparison to alternatives such as a split which requires both process to run at the same time. The modular SSMG used in RECLAIM*Lite* allows for the system to be replaced by a larger microwave architecture proposed by ILUMEN while retaining the existing chamber selection. The use of the modular SSMG and microwave switch enable efficient control of the system and ease of use for users.

[fortnite 7/20] also clarify if this will be in the full system or a different switch/system will be used instead **[wetz 7/21]** I agree, but to what extent do we want our entire report to revolve around "Hey we built this, but if we had a lot of money I promise you would like it even more." Is our report just explaining how RECLAIM*Lite* is a proof of concept for the actual RECLAIM system? Should we be explaining a general design rationale for the RECLAIM system that would be built or what we have actually designed for this

competition?

Many of the microwave components used in the RECLAIM*Lite* system have different sized waveguides, which requires adapters to interconnect them. These adapters not only increase the size and mass of the RECLAIM*Lite* system, but also can contribute to insertion loss and reflected power. These component availability constraints won't pertain to the future RECLAIM system, meaning all components would use the same waveguide size. This will reduce the mass and size of the system while also limiting losses caused by impedance mismatch between components. **[7/21 wetz]** Bilen used this as example of what we should be trying to convey in report. RECLAIM*Lite* follows the idea of RECLAIM but is the great value version.

Water cooling is used for the SSMG, isolator, dummy load, and tuner cooling jacket to maintain safe operating temperatures. The tuner incorporated into the RECLAIM*Lite* system was previously used in the milestone system with a 1.5 kW Daihen ATM15A magnetron. However, RECLAIM*Lite* is designed to operate at a maximum power output of 3.0 kW. At this higher power, the heat generated will be substantially higher, which can affect the performance of the tuner. The cooling jacket removes heat from the tuner body during operation, mitigating this effect. This allows the tuner to maintain an operable temperature while continuing to perform its function during smelting or pyrolysis.**[7/21 wetz]** Should information about the cooling jacket be mentioned since it doesn't pertain to the actual RECLAIM system. What emphasis are we putting on explaining things in our competition system that won't be in the future RECLAIM system. This will be even more essential for a future RECLAIM system that could utilize the 6.0 kW capabilities of the SSMG. The dummy load and the circulator are crucial for the protection of the SSMG from reflected microwave power. As the redirected waves from the circulator are absorbed and turned into heat by the dummy load, the cooling water removes the heat generated in the dummy load to regulate the temperature. The cooling loop improves the reliability of the components by preventing an excessive increase in temperature during extended operation. Water cooling is used because it allows for a compact design that can fit within the dimensions of the BLiSS rack. **[wetz 7/20]** Charles maybe add onto this for platics cooling

2.1.1.2. SMELT Subsystem Design

- Drip cast explanation and how it is applied .
- Mullite tube straight into crucible explanation
- Casting explanation and purpose for drawer being welded to the bottom of the chamber
- Purpose of boron nitride being sprayed on the crucible in relation to reduction of slag
- Purpose of thermocouples to obtain thermal data of the system
- Address hazard mitigation strategies used to create the final product of the SMELT system
- Address the importance of specific alignment of the sprue and riser holes in relation to the distance from the drag.

SMELT is built around a stainless steel resonance chamber, a Silicon Carbide (SiC) crucible, a sand-cast mold drawer, and a variable concrete vibrator. SMELT utilizes a continuous casting methodology to allow for minimal user interaction with high temperature materials. The continuous casting system is enabled by a conical crucible design. An extrusion is designed to allow flow once the aluminum reaches its melting point while not permitting solid aluminum to pass through. This approach produces a more repeatable fill behavior due to a standardized flow rate from the set geometry of the crucible. The resonance chamber is

sealed using a bayonet mount design to lock the chamber vertically once the user twists it clockwise. The lid was designed to prevent any microwave leakage from the top and to allow ease of access to the rest of the system for maintenance purposes. Thermocouples are attached to the lid and the base of the chamber to obtain internal heating values within the SMELT chamber. **[Parker 7/21]** This whole paragraph needs to be rewritten, some sentences are so short to the point they can be combined, specifically the middle group and the thermocouple mention should be introduced in a better way

Before aluminum shreds are introduced, the SMELT subsystem is preheated for 5 minutes at 1kW to avoid cold spot formation within the crucible, which could otherwise reduce melt uniformity. The aluminum feedstock is pre-processed through a VEVOR shredder, producing smaller pieces that increase the amount of material that can be loaded into the crucible while reducing melting time. The modified aluminum is then fed through a mullite tube to fall into the crucible. The mullite tube itself is attached to the bayonet lid and extends into the crucible, creating a dedicated feed channel. A hole matching the tube's outer diameter is machined into the center of the bayonet lid where the mullite tube is secured using high-temperature refractory furnace epoxy. Mullite was selected for RECLAIM*Lite* because it is a microwave transparent ceramic and resistant to extreme temperatures. The aluminum is then heated to its melting point to allow flow through the 8 mm exit and automatically filling the sand mold geometries to create a reliable part.

[Parker 7/21] The 2 paragraphs talking about the bayonet mount and the mullite tubing should be combined to be shorter while keeping in tact the information being talked about and the crucible molds figure needs to be mentioned in the text prior to showing it.

SiC was chosen for the crucible because of its dielectric properties and its ability to couple well with 2.45 GHz. Specifically, the crucible is made up of a mixture of 51% silicon carbide (60-90 grit), 34% silicon carbide (400-600 grit), and 15% sodium silicate. For the crucible making process, it is packed into a 3D printed mold pictured in Figure 4 where it dries before being sintered in a muffler furnace. The crucible's interior features a conical bottom geometry to optimize molten aluminum's flow towards the crucible's orifice. The crucible interior is coated with a layer of boron nitride to create a non-wetting barrier between the crucible wall and melt preventing slag buildup across multiple cycles.

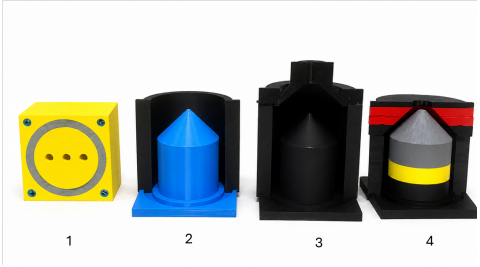


Figure 4: Iterative mold making process for crucible manufacturing

A variable concrete vibrator mounted on supportive 80 20 aluminum frame uses mechanical vibration through the chamber walls in the crucible to disrupt the formation of aluminum oxide. By continually agitating the boundary layer at the crucible and it's orifice (Oliveira et al., 2021), the vibration ensures there is no freeze plug behavior at the hole in the crucible and encourages flow.

To operate SMELT safely, certain personal protective equipment is employed to use components of the design. Heat protective gloves, a heat resistant apron and a heat shield are used to protect users from harsh temperatures and superheated alloys. A N95 respirator and gloves are used to handle the ilmenite sand mold. **[Parker 7/21]** Need to talk with others to see if the hazard mitigation is needed, ik I put it in but this information is already talked about later and also mentioned previously in section 1. Also the crucible paragraph, if possible, should be shortened.

2.1.1.3. Plastics Subsystem Design

- Overall reaction chamber design to support an IR pyrometer, vacuum gate valve, pressure sensor, and gas-in gas-out
- Double-pass microwave absorption technique with variable-frequency microwave generation for homogenous heating solution
- Hopper that automatically opens and closes to accept feedstock in the system
- Cooling loop to condense heavier fraction oils / waxes into vac-trap
- Double diaphragm, hermetically sealed LABOPORT pump to move hydrocarbons through system
- Venting / refill solution that lunar-deferred RECLAIM will utilize the vacuum of space with for venting / depressurization
- Address safety hazards present with toxic gases and liquids that could be produced — mention how nitrogen-bearing plastics are dangerous due to cyanide

The plastic pyrolysis subsystem is designed to decompose plastics, specifically focusing on breaking down long form hydrocarbons into shorter-chain, high-value molecules—such as synthetic oils, combustible gases, and solid carbon char via random chain scission of bonds.

Plastic pyrolysis is a highly complicated chemical process that must undergo strict safety considerations in order to ensure the prevention of toxic gas leakage, microwave leakage, and human interaction with hot surfaces. Many considerations were made in the development on the subsystem in order to ensure crew safety in an eventual lunar integration. **[fortnite 7/16]** what are those considerations?

The pyrolysis system features a reaction chamber, a distillation unit, and a filtering system that separates pyrolysis oil from gaseous hydrocarbons. A multi-phase pump system is used to simulate the vacuum of space, as well as collect hydrogen-rich syngas. The distillation system incoroporates itself with the same water-cooling supply that the microwaves use to reduce complexity of the cooling system. **[charles 7/16]** mention vacuum-assisted pyrolysis earlier in the paper - will need to confirm products are "hydrogen rich"

The reaction chamber is a retrofitted ISO200 full nipple used to connect wide-diameter vacuum chambers. It serves as the body of the reactor. It contains the heat, microwaves, and pyrolysis products present.

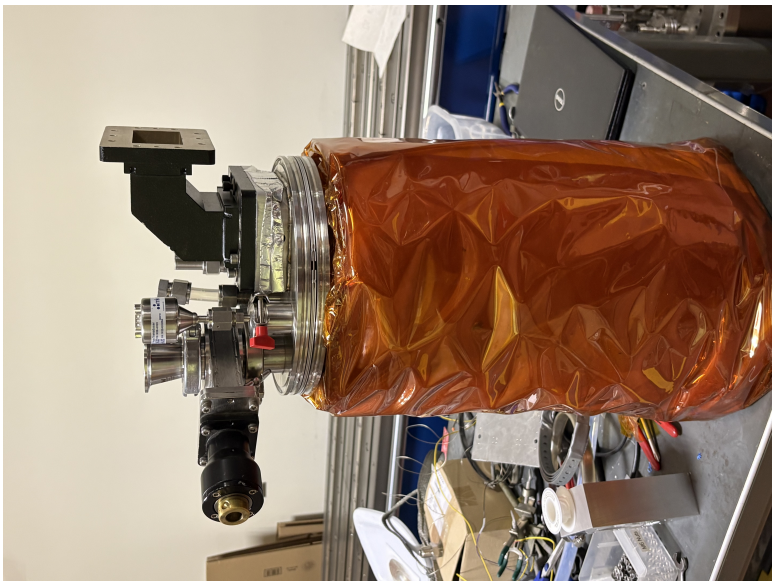


*Figure 5: RECLAIM Reactor Vessel Assembly **[charles 7/16]** will fix idk why its sideways - probably will get a nicer picture too - isometric or something like that - maybe compare with CAD?*

The body of the reactor is housed in ~5/4" **[charles 7/16]** confirm insulation thickness on top and bottom thick ceramic insulation fiber to contain the heat of the system. It is then wrapped in ~3 layers of Kapton wrap & tape in order to prevent ceramic fiber from being touched during handling, as well as to adhere the insulation to the chamber. The insulation acts like a sleeve and can be removed for long term maintenance.

The top of the reactor houses a removable ISO200 flange that was machined to house several critical components. A WR340 waveguide sits above the fused quartz window and is positioned between a silicon gasket with a water-cooled channel to ensure the quartz does not heat up during operation, to prevent degradation. A vacuum gate valve that opens to allow the insertion of feedstock material into the reactor. A KF50 viewport to allow a non contact IR sensor imaging of the surface of the SiC bed. The non contact IR sensor is interchangeable, with a 1:2 optics version being available for a temperature-averaged measurement, and a 1:75 optics version available for hot-spot monitoring. The flange also features a 902B Piezoelectric pressure transducer to precisely monitor the pressure inside of the chamber. The piezoelectric diaphragm was chosen because it is compatible with pyrolysis gas service, as the internal electronics are protected from the gas stream via a physical barrier between the sensors and the pyrolysis chamber. **[charles 7/16]** confirm details on instruments and assumptions on gas safety for piezo transducer - it may experience gunking over time - need to confirm through testing

To connect the reactor to the distillation and condensing unit, a 3/4" braided steel hose is connected to Swagelok **[fortnite 7/17]** is swagelok cpaitalized? bulkhead on top of the chamber for ease of removal. It is then connected to a 3" x 18" tri-clamp baffled condenser often used in food service applications **[fortnite 7/20]** is there a model name instead? the food service thing sounds a little out of place. It is angled at a 45 degree angle to allow gravity to collect any condensed waxes and oils the system will produce.

A retrofitted cold trap is then used to rapidly increase the volume of the gas to condense out any condensate still suspended in the gas stream. It then flows down into a butterfly valve that can be opened to collect the oil and wax products **[charles 7/16]** will need to confirm that this works and that a vac-trap isn't necessary

Both the cold trap and the condenser receive cooled water through the water-cooling loop that RECLAIM also uses to cool the microwave system.**[charles 7/16]** need to confirm if room-temp water will be accept-

able through final testing This system features a quick disconnect subsystem to easily drain and purge the cooling system for maintenance.

Non-condensable, light hydrocarbons pass through the condensation section and are transported further downstream. A coalescing oil filter with **[charles]** confirm size of particles that oil filter catches a 5 micron filter serves as a final barrier for catching oil. Past the oil filter sits a plumbing series used to switch between the pre-process and purge ball valves used to control the output direction of the system. A second 902B piezoelectric pressure transducer is mounted to measure the pressure drop across the filter and distillation units, which is used to monitor the wax buildup that happens over time in the system.

Past the plumbing tees is a Laboport KNF 840 pump. **[charles 7/16]** again, confirm name and details of pump and instrumentation we are using. **[Noah 7/21]** I believe it's a LABOPORT UN 820.3 FT.40PThe Laboport pump is a laboratory grade double-diaphragm pump meant to carry saturated vapors. It features PTFE diaphragms and housing that is chemically inert to hydrocarbons.

Many design decisions were made in order to prevent the combustion of hydrocarbons within the system, as compressing hydrocarbons presents a critcal safety hazard to crew. Oxygen ingress and displacement is essential in maintaining a strict limit of oxygen within the system. **[fortnite 7/17]** what is that limit

The sizing of the pump does not allow the system to stay under the minimum combustion pressure for flame propagation limits due to the nature of pump dynamics. The high volume of gas being produced within a small time window along with the high differential **[fortnite 7/17]** differential of what the pump must move fluid across creates this impossibility. An exceedingly over-sized **[fortnite 7/17]** calrify over sized. do you just mean bigger? more pwoerful? pump would be necessary to operate vacuum pyrolysis at high volumes. To account for these high volumes overpowering the lab pump, an overpressure line is attached to the output utilizing a check valve. If the system ever exceeds atmospheric pressure relative to the lab setting, the check valve will open, and the pyrolysis products and purge gas are released. **[fortnite 7/17]** released to where? clarify that gases are not being released into the lab

There is also an inherent combustion hazard in this system due to hydrocarbon production. To mitigate this, an inert gas is inserted into the system which acts as a carrier gas for pyrolysis products while also displacing oxygen that enters the system. **[fortnite 7/20]** extremely similar to the one two paragraphs ago. probably just combine the two or only keep one

Two operating modes exist for this inert purge: Positive pressure and negative pressure. The positive pressure operating mode involves venting to space, closing the valve, and filling the system with an inert gas until the check valve begins leaking out. This ensures zero oxygen is entering the system by leaking inert gas out instead of in. This operating mode is inherently safer, but also using significantly more inert gas, which is a consumable that would need to be recycled out of the syngas. The negative pressure operating mode is more efficient, but presents more risks in operation. **[charles 7/17]** I'm not sure if it's worth explaining the operating modes more or just the entire pyrolysis system as a whole. The safety mitigation is directly tied into the operating mode so I think I have to include it - also, clarify the existence of the two pumps, one for vacuum representing space and the other collecting products - **[charles 7/17]** I will move the operating procedure to 2.2 instead of 2.1

To operate the system in the negative pressure configuration, the system is evacuated utilizing the purge pump, and pyrolysis is started under vacuum conditions. This changes the operating chemistry of the system **[charles 7/17]** expand as soon as testing and operating of negative pressure system is finished

The minimum oxygen concentration for combustion of hydrogen, which is the lowest M.O.C product produced by the reactor, is 5.0% by volume. The purge gas is thus used, along with the vacuum tight seals, to keep oxygen concentration below by measuring the leak rate and volumetric production of hydrocarbons via

a flow meter. A factor of safety of 2.0 is used, so the oxygen concentration within the system never exceeds 2.5%. **[charles 7/17]** confirm factor of safety and numbers here

After the completion of pyrolysis, the system is pumped down via the laboratory pump to collect as much product gas as possible. The system is then pressuzied, and the liquid products are collected via the vac-trap.

2.1.1.4. Automation

- Explanation of stepper motor incorporation — feeding both chambers with a sliding rack
- HMI that astronauts can interact with and get real-time feedback from system

To ensure precise control and minimize crew interaction, the RECLAIM system utilizes a control state machine automated by which its states are driven by a National Instrument cRIO **[julian 7/20]** the cRio is a data aquisition system. albeit, cRio should be well known. Also don't believe we are still using controllino **[Noah 7/21]** Lucas change this first sentence based off sensor inputs, system process, and product output to drive an optimal system process. RECLAIM utilizes the LabVIEW program on the cRIO to read all sensor input and allow the user to select the process of their choosing without further human interaction.

The automation subsystem features a stepper motor assembly responsible for operating the gate valve at the top of the pyrolysis chamber. The assembly consists of a stepper motor coupled to a custom mounting hub, which is directly bolted to the gate valve's wheel. This configuration allows the system to accurately control the opening and closing of the valve to manage feedstock intake. To facilitate material access, a secondary mechanism utilizes two stepper motor-driven actuators to control a sliding door on the top of the chamber. These actuators allow for controlled, automated access to the internal chamber for maintenance or material loading.

Each movement is synchronized with the current system state to ensure safety and prevent interference with the vacuum or thermal integrity of the chamber. This prevents starting a process too early that can cause any faults that would be deemed inefficient or unsafe.

Beyond mechanical actuation, the automation subsystem integrates an HMI that provides astronauts with real-time feedback on system status, actuator positioning, and process telemetry all based off the sensor and system readings from the cRIO. This interface ensures that operators can monitor the process remotely and intervene if necessary, reducing overall crew workload and increasing mission safety.

All telemetry and system data will be fed back to the Digital Twin to work with real time data and system flow down. This allows calibration with the Digital Twin and get better simulation results **[Lucas 7/16]** Luke I would expand upon this and attempt to explain the digital twin better. **[Noah 7/21]** I don't understand the digital twin enough to expand on this, Lucas... Attack

2.1.2. Performance Analysis

Req: Preliminary evidence and analysis predicting performance including mass, volume, and power requirements. (This is predictive testing; actual test results are in 2.2.)

2.1.2.1. Pyrolysis

- Heat exchange predictive performance
- Microwave reflectivity / absorption predictive performance
- COMSOL analysis

The plastics system underwent in-depth analysis using COMSOL multiphysics in order to calculate the ideal thermal and dielectric properties of the system.

The goal of the testing was to achieve minimal hot spots, maximal microwave coupling, and minimize bed height. These test variables achieve separate goals that were verified later via testing.

The top-down feed of the microwave waveguide is atypical for cylindrical resonance cavity geometries, but was necessary to maintain sufficient S_{11} **[ammar 7/21]** what is coupling in the resonance cavity. The silicon carbide bed must be deep enough to allow the full absorption of the microwaves, but not be too thick as to exaggerate hot spotting and require more energy to heat up. Silicon carbide's specific heat is high, as it a refractory ceramic. **[ammar 7/21]** feels like this last sentence should go in the next paragraph

Silicon carbide's loss tangent increases exponentially as function of temperature, thus the ideal bed height to allow the double-pass of microwaves to be sufficiently absorbed changes as the bed heats up. An average bed height must be selected as to trade off reflections, hot spotting, and thermal loading.

Hot-spotting presents a critical issue to the system due to the positive feedback loop present due to silicon carbides dielectric properties. To mitigate this issue, the variable frequency band of the SSMG is utilized to change the standing wave resonance of microwaves within the chamber. This changes the heating pattern within the bed, thus replicating the effect of mechanical agitation that is often necessary in microwave engineering due to distinct hot-spots. This effect can be seem in Figure 6 **[wetz 7/20]** use hotspotting multiple times in this section make sure not sure if it is a real word

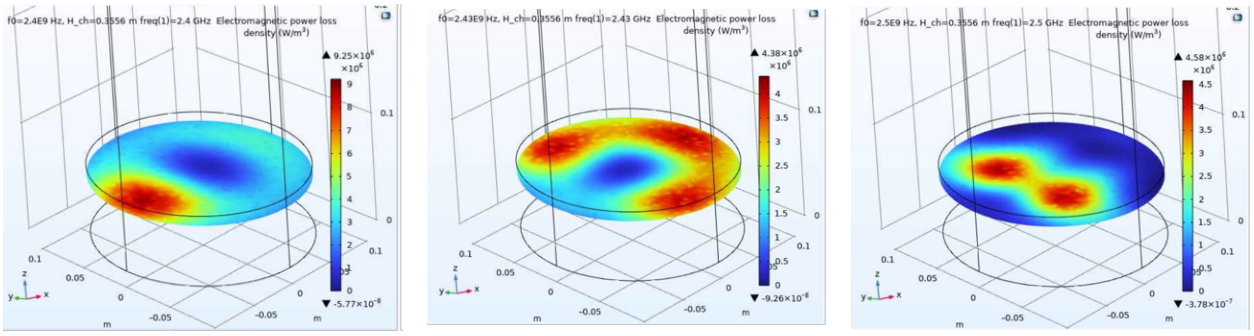


Figure 6: Simulated electromagnetic power loss density distribution within the cavity at 2.50 GHz, 2.43 GHz, and 2.40 GHz (left to right), showing hot-spot location and intensity shifting across the frequency sweep.

As the chamber heats up, the feedstock will begin to degrade under the inert environment. Polyethylene plastic is comprised of repeating units of ethylene (C_2H_4) that form extremely long hydrocarbon chains. These long hydrocarbon chains act as the starting material for the pyrolysis process. The reaction begins when heat is added to these hydrocarbon chains. The heat causes chain scission reactions to occur, leading to the degradation of these long chains. C–H bonds tend to break first, followed by the C–C bonds as the temperature increases. The chains continue to degrade further as the reaction progresses until volatile gases are formed [5] **[charles 7/19]** make sure to re-include reference here from old report. Volatile gas products

that are produced during this process include hydrogen gas (H_2), methane (CH_4), ethylene (C_2H_4), propene (C_3H_6), etc. These volatile gases form via reactions with radicals and small hydrocarbon compounds after the chains have degraded. The highly reactive unpaired electrons in the radicals make these reactions possible. These chemical reactions are provided in Figure 7. **[charles 7/19]** this is copy and pasted from the first report, I think it does a good job of explaining the chemistry behind polyethylene decomposition

$$\text{Polyethylene Carbon Chain} \xrightarrow[\text{(Heat)}]{\Delta} \begin{array}{l} \mathrm{C_2H_4 + CH_3^\bullet} \\ \mathrm{C_3H_8 + CH_3^\bullet} \\ \mathrm{C_2H_5 + H^\bullet} \\ \mathrm{C_3H_7^\bullet} \end{array}$$

$$\begin{aligned} \mathrm{C_2H_4 + CH_3^\bullet} &\longrightarrow \mathrm{CH_4 + C_2H_3^\bullet} \\ \mathrm{C_3H_8 + CH_3^\bullet} &\longrightarrow \mathrm{CH_4 + C_3H_7^\bullet} \\ \mathrm{C_2H_5^\bullet + H^\bullet} &\longrightarrow \mathrm{C_2H_4 + H_2} \\ \mathrm{C_3H_7^\bullet} &\longrightarrow \mathrm{C_3H_6 + H^\bullet} \end{aligned}$$

Figure 7: Chain scission and radical reactions during polyethylene pyrolysis.

The pyrolysis is also suited to break down other long chain bonds under an inert environment such as celluose, as found in cotton. Bio-oil is a focused area of research dedicated to the recovery of fuels through biomass. **[charles 7/19]** will need to expand on chemistry here if we end up doing shirt

To cool down the pyrolysis products to protect downstream hardware as well as condense oils and wax, a bench-top distillation unit cools down the gas stream. **[fortnite 7/20]** i'd restructure this setence to: "A bench top distillation unit cools down the pyrolysis products in order to protect the downstream hardware." but that also is very short to be its own paragraph, could be be used somehwere else instead

In order to estimate the amount of energy that needs to be dissipated from the pyrolysis products, several factors need to be taken into account such as the heat transfer coefficient, the nominal flow rate, the desired temperature, and the latent heat of condensation. An analytical solution to this is difficult due to the wide speciation of pyrolysis products. **[fortnite 7/20]** so then how did you do it? this is an important factor so make sure there is some explanation behind your design, even if it is just restating the comsol models**[ammar 7/21]** this paragraph is kinda out of the blue, we've started discussing the chemistry and processes at this point. it could be earlier in the section

The goal of the distillation unit is to condense the lightest product that is liquid at room temperature, so the liquid-gas products can be properly separated and stored. The lightest species in this case is pentane, thus it served the basis of required energy removal. **[charles]** I could go way more in-depth with the derivation for the heat exchange and all of that. They explicitly ask for thermal data, I'm not sure how in depth I need to be here. fortnite-this relates to my previous comment, i think just have a little more in depth (not equations or anything) just to show that the proper care was taken

The rapid increase in volume after the baffled condenser into the cold trap also assists in condensing vapors out of the gas stream through adiabatic cooling principles. This is later assisted by the oil filter, which acts as a final catch for condensable vapors.

2.1.2.2. SMELT

Before testing, computational models and calculations were derived based on known values from the SMELT Team, 2023, as well as target values for the system. To calculate the approximate amount of time for the SMELT subsystem to run one batch based on a specific input power, we use

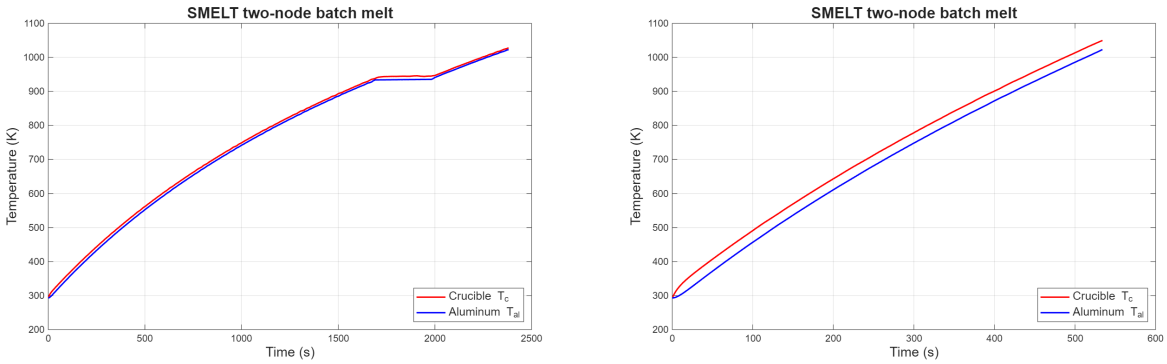
$$m_{\mathrm{c}}\, c_{\mathrm{p,c}}(T_{\mathrm{c}})\, \frac{dT_{\mathrm{c}}}{dt} = \eta P - \dot{Q}_{c \to Al} - \dot{Q}_{\mathrm{loss}} \tag{1}$$

and

$$m_{\mathrm{al}}\, c^{\mathrm{eff}}_{\mathrm{p,al}}(T_{\mathrm{al}})\, \frac{dT_{\mathrm{al}}}{dt} = \eta P - \dot{Q}_{\mathrm{c \to Al}}, \tag{2}$$

where m is the respective masses for the two main parts of the system (subscript c is control volume and al is for the aluminum), c_{p} are the respective specific heat coefficients, η is the absorptive efficiency of the crucible and the insulation, P is the power input into the system, $\dot{Q}_{\mathrm{c \to Al}}$ is the heat the crucible conducts to the aluminum, and $\dot{Q}_{\mathrm{loss}}$ is the heat lost during the run process. Equation 1 presents the crucible and insulation as its own control volume with differing temperature changes compared to the aluminum, which is presented in Equation 2. Whereas both are functions of time and temperature, they both are based on relative values to the respective base material covered.

Two differential equations can be derived using Equations 1 and 2. They are based on time and temperature and allow for some variables to be filled in based on known constants such as mass, specific heat, power, and absorptive efficiency. The computations of exact temperatures at the end of the process once the aluminum is flowing into the sand mold were calculated in MATLAB to obtain an estimated time it takes for one batch to be melted in the crucible and flow into the sand mold as shown in figure 9.



(a) 1 kW *(b) 3 kW*

Figure 8: Time-to-melt results at two power levels.

The most noticeable differences between the two different graphs is the time it takes for the metal to reach 1023 K, slightly higher than the melting point of aluminum (933 K). During Earth-based testing, the aluminum charge is heated in two stages. Early in a run, before the charge makes good thermal contact with the crucible, radiation from the inner surface of the crucible dominates. As the crucible passes the melting point of aluminum and the charge begins to come into complete contact with the wall, conduction takes over and delivers most of the remaining sensible and latent heat. Convection contributes no heat to the charge and acts strictly as a loss.

In a lunar environment, the absence of an atmosphere eliminates convection and gas-phase conduction, leaving solid-contact conduction and radiation as the active thermal paths. Radiation is the governing surface

loss parameter due to the loss channels of convection being gone and the crucible being surrounded by the cold vacuum of space which causes an increase in radiative loss compared to Earth based model.

Figure 9 only gives thermal data and how long it will take for the aluminum to be heated to the proper temperature and they do not include the time it takes for it to flow into the sand mold. The following figures will provide a more detailed flow analysis of the molten aluminum as it flows from the crucible into the sand mold, shown in Figure 9.

The graphs in Figure 10 display the time it takes for the aluminum to leave the crucible and enter the sand mold once it has passed 933 K (660°C). The top graph represents the aluminum as it transitions from solid to liquid over a total time of approximately 190 seconds. The bottom graph compares the crucible temperature to the temperature of the liquid aluminum pool that forms as the aluminum melts. The pool remains at a constant 933 K because the aluminum drains from the crucible into the sand mold as soon as it melts, leaving no time for the liquid to superheat, while the crucible temperature continues to increase. The plateau that occurs once the crucible reaches 980 K is solely because of the model. This model holds the pool temperature at 933 K, the crucible only needs to sit a fixed increment above the melting point to sustain the melt, so it stalls at 980 K. In a real experiment, the crucible temperature would continue increasing until it plateaus at the actual maximum temperature it can absorb, governed by the balance between absorbed microwave power and heat loss. This model does not include freeze-off scenarios, and the crucible absorptive efficiency,like the graphs in Figure 9,is based on the absorptive efficiency reported in the SMELT report.

Thermal data for the SMELT system is further shown through the aid of COMSOL which was used to understand the thermal properties of the completed chamber prior to being able to run tests. The COMSOL models in figure 9 show the temperature of the chamber itself as well as the attached drawer and the crucible when the system is at the melting point of aluminum.

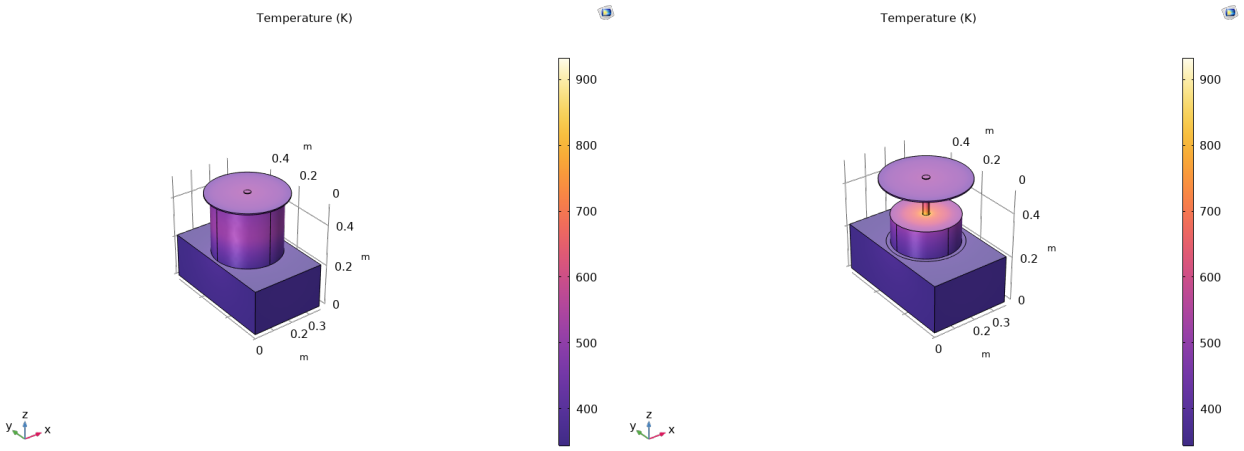
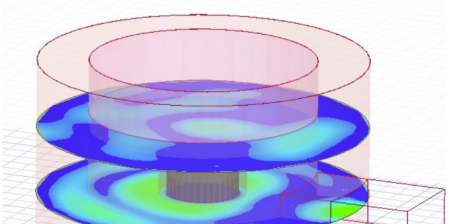


Figure 10: COMSOL physical models of the chamber thermal properties and the dispersion of heat around the chamber and drawer system.

The COMSOL models help to understand the heat loss that occurs within the system as the aluminum melts. The data gives reason to surround the chamber and drawer with insulation to keep the temperature as high as possible. This allows the continuous pour system to be feasible and mitigate the chances of freeze-off as the molten aluminum flows from the crucible to the sand mold.

Utilizing an ANSYS HFSS electromagnetic model (SMELT Team, 2023) allows for better understanding of the high intensity regions



within the SMELT chamber and the importance of placement to minimize reflection losses. The mounting of the WR340 waveguide in the middle of the resonance chamber allows heat to be concentrated around the center which is crucial to the smelting process. The simulation also helps optimize crucible placement and how efficient our chamber is at assisting the flow of microwaves to heat up the crucible.

2.1.3. System and Component Specifications

Req: System-level and component-level design specifications for hardware and software, including bill of materials and system mechanical schematic. This section should also include the maximum allowable temperature (°C) and maximum allowable working pressure (PSIG) for the equipment. (We will have a BOM for every schematic; anything not covered in said schematics goes to 2.4 — per UA feedback.)

- (Waiting to hear back from UA about how this section is different than 2.3 and 2.4)

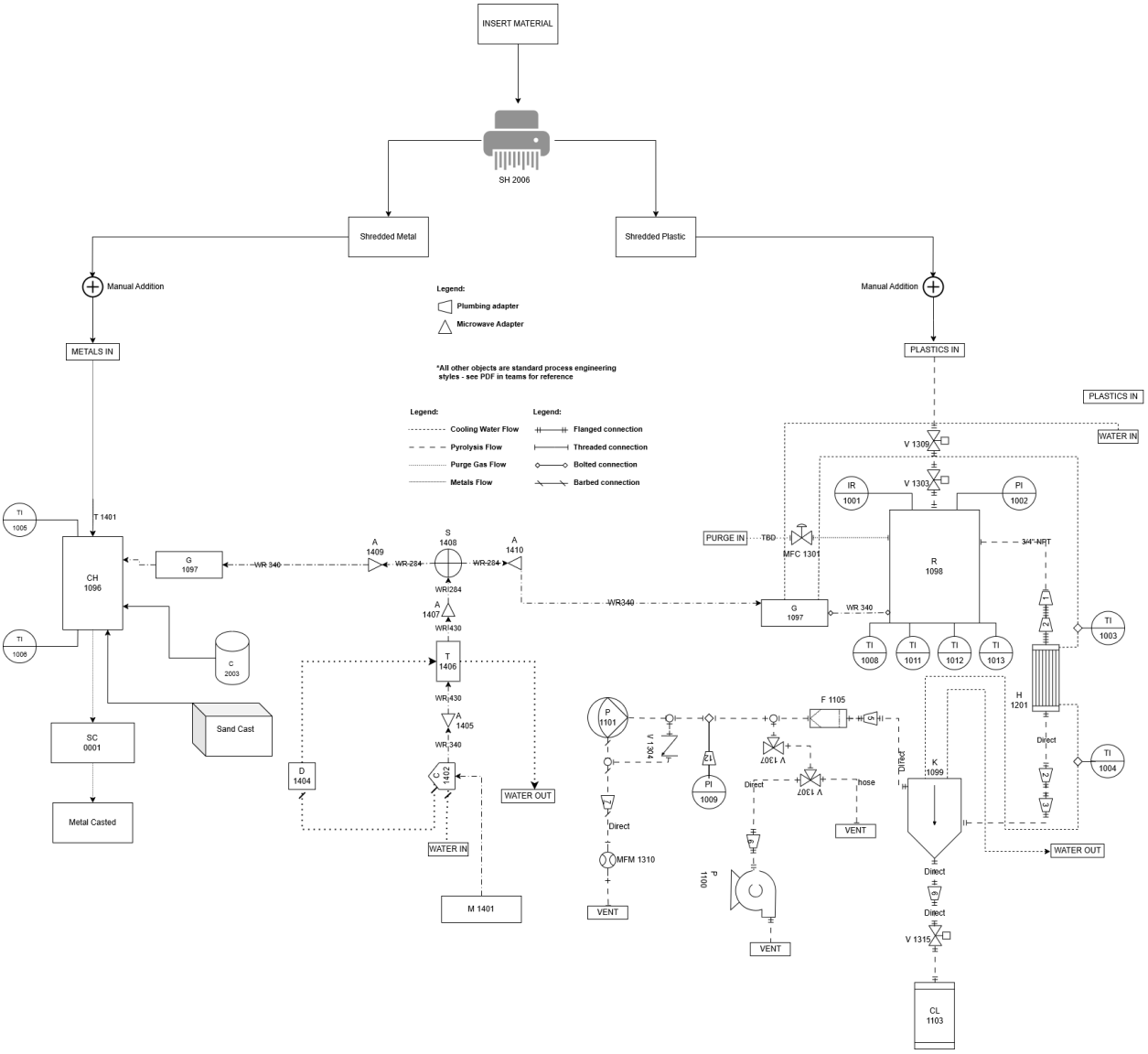


Figure 12: Draft of P&ID ***[julian 7/20]*** *I cant read this.* ***[charles 7/20]*** *might have to remove - I think it's nice that it shows the overarching systems, but you can't read it - not sure about this*

Pyrolysis Subsystem Equipment Table

Description of Equipment and Supplier	P&ID Ref	Estimated mass (kg)	Estimated volume (cm^3)	Maximum allowable working pressure (PSIG)	Maximum allowable operating temperature (°C)
Pyrolysis Reactor (custom, R-1098)	R-1098	TBD	TBD	TBD	TBD
Knockout Pot (Swagelok K-1099)	K-1099	TBD	TBD	TBD	TBD
KNF LABOPORT UN840.3 FTP Pump	P-1101	TBD	TBD	TBD	TBD
Spiral Coil Condenser (2")	C-1102	TBD	TBD	TBD	TBD
Haskel Booster	H-1201	TBD	TBD	TBD	TBD
Spillback Back-Pressure Regulator	PCV-1310	TBD	TBD	TBD	TBD

2.2. Final Round Operations, Analyses, and Test Results

Recommended length for 2.2: 6–8 pages

2.2.1. Concept of Operations

Req: Concept of operations for one cycle and any related analyses or testing, including thermal analysis and structural analysis, if relevant to your solution.

RECLAIM's operations begin with the shared preparation and startup procedures for both processes. Both processes will complete 4 steps before microwave power is turned on to ensure safe and successful completion of the desired process. Once the SSMG is turned on, the system branches off into two separate paths for plastic pyrolysis and aluminum smelting. Once the desire process is completed, both processes follow the same shutdown procedures. Finally, the products from the chosen process are recovered as depicted in figure 13.

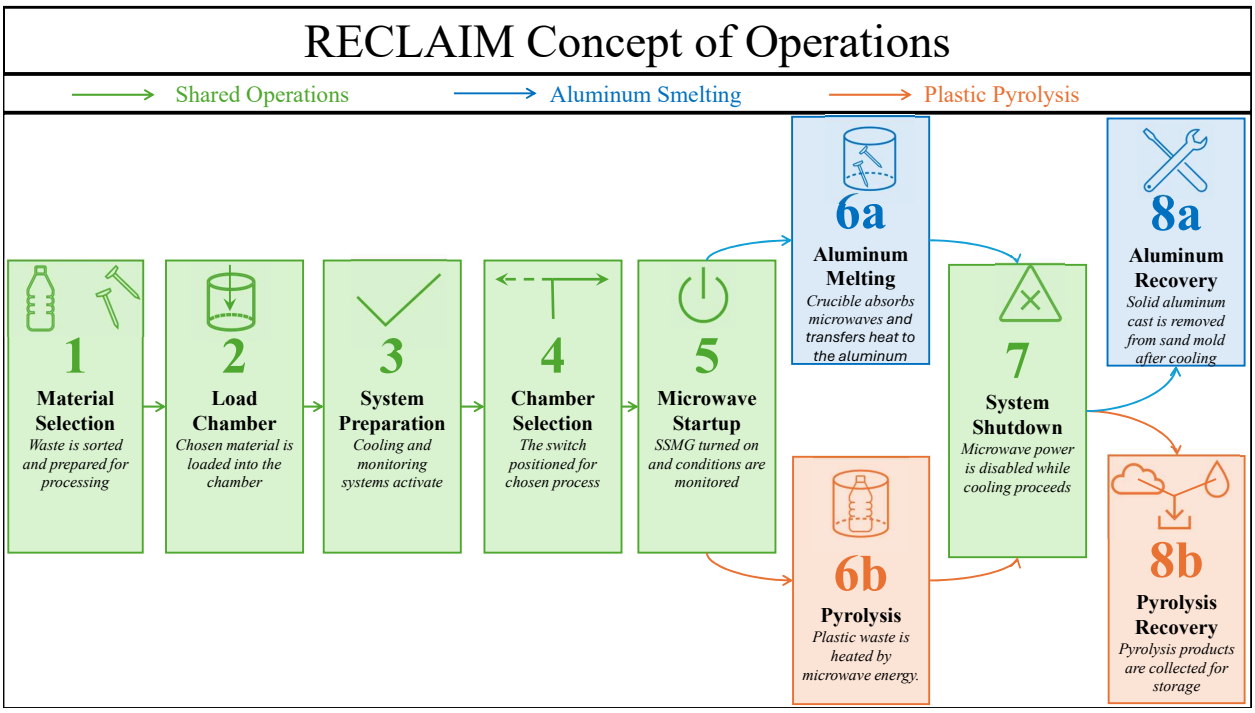


Figure 13: RECLAIM Concept of Operation

2.2.1.1. Microwave Testing

- Component Integration Testing
- Cooling Jacket Endurance/Temperature Test
- Power Sensor Comparison Test

Testing of each microwave component was completed sequentially from the isolator to the switch to verify all components functioned properly prior to full system integration. The isolator and dummy load were tested first to ensure reflected microwave power was redirected into the dummy load instead of returning to the SSMG. The tuner was then tested to ensure reflected power was minimized, while forward power to the selected chamber was maximized. Testing of the tuner also included confirming that the controller gave accurate readings of forward and reflected microwave power. Finally, the switch was tested to ensure microwave power was routed into the selected chamber with minimal RF leakage from the opposing side.

Coolant flow testing was performed to ensure nominal conditions. The tuner is the only component not rated for operation at 3.0 kW, additional testing was done on the cooling jacket to ensure it maintained an operable temperature for the tuner. An Extech I5 Infrared Camera was used to measure the temperature of the cooling

jacket every 5 minutes for 40 minutes at 3.0 kW of forward microwave power setting. The results showed that the cooling jacket's temperature increased from 22°C to 29°C during the first five minutes and then remained between approximately 29°C and 30°C for the remainder of the test.

The sensors built into the tuner are limited to 1.8 kW, which due to power losses is about 2.0 kW of SSMG power setting. Since the RECLAIM system operates at 3.0 kW, testing was done to verify the tuner's functionality past the sensor's limit. A Mini Circuits USB 8GHS power sensor was connected to a Gerling GL206 dual directional coupler added after the tuner to measure forward and reflected microwave power. The coupler samples a small quantity of the microwave power flow as it moves through the waveguide, and the power sensor reports it in decibel-milliwatts (dBm). The values are then converted to watts using the coupler's calibration factor. The SSMG's power setting was increased in intervals of 0.5 kW, starting at 0.0 kW and ending at 3.0 kW. Forward and reflected microwave power were recorded from the coupler's sensors and compared with the data displayed by the tuner's controller. Both readings stayed within 3.5% of each other while inside the tuner's readable measurement range. A line of best fit was created from the collected tuner data from 0.0 to 2.0 kW of SSMG power setting to predict the tuner's expected outputs between 2.0 kW to 3.0 kW. This regression line for the predicted tuner response is displayed in figure 14 as a blue dashed line. The coupler's independent readings for 2.5 kW and 3.0 kW of SSMG power setting were within 2.0% of the predicted tuner's response from the line of best fit. In addition, the reflected microwave power remained less than or equal to 2 W throughout testing for both sensors, indicating the tuner is performing its function to limit reflected microwave power. Based on the testing, the automatic tuner demonstrated proper functionality during operation for up to 3.0 kW of SSMG power setting. **[wetz 7/21]** All of the testing done for microwaves has nothing to do with actual RECLAIM system other than maybe preliminary component verification in 1st paragraph. Should this section be chopped or shortened if we are trying to avoid addressing RECLAIM*Lite*'s limitations. Bilen made comment about this section in particular

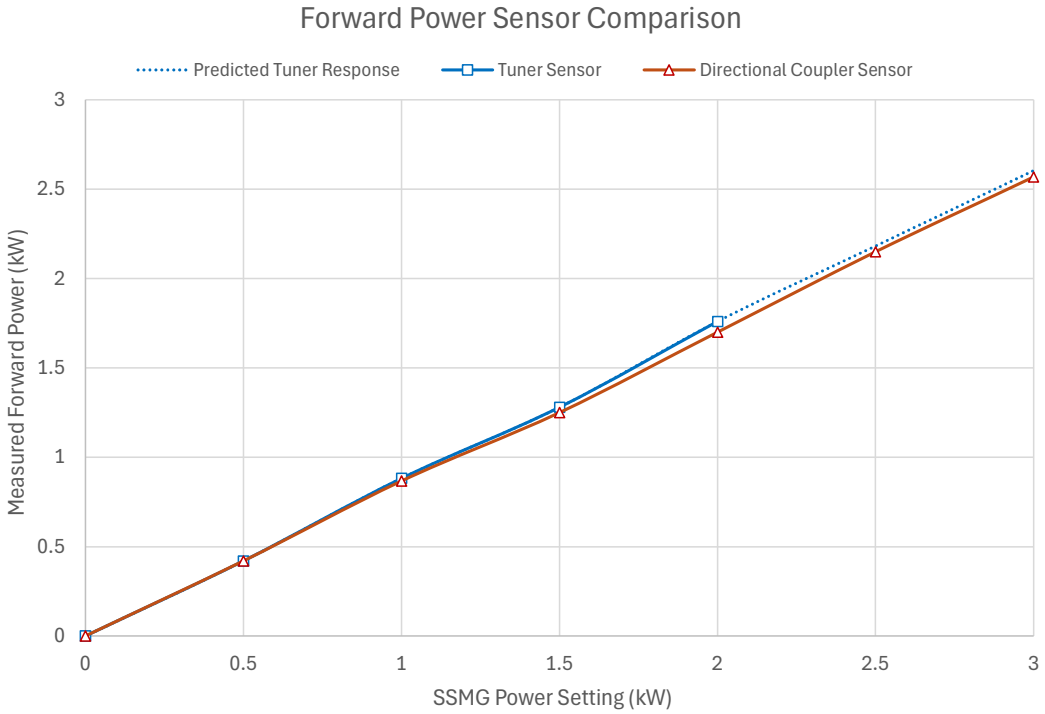


Figure 14: Comparison of measured forward power from tuner and coupler sensors.

2.2.1.2. SMELT Testing

- Proper PPE being equipped and wrapping the crucible correctly in insulation
- Proper alignment of both the crucible to the chamber hole and the sand mold to the chamber hole
- Check for microwave leakage while testing

For preparation of the ilmenite sand mold, an N95 respirator and nitrile or neoprene gloves are required due to the fine abrasive of the ilmenite and the particulate matter generated when handling and breaking apart the dried ilmenite mold material.

When running the system, operators should wear thermally safe working gloves that reach the elbows, and a face shield to protect against potential splash hazards. A second user must be readily available with the same PPE to assist the primary handler when necessary.

The primary user can estimate the total input mass needed through knowledge of the patterns density, volume, and mass as well as the density of aluminum. If the pattern was 3D printed all that is needed to properly calculate the input mass is the volume of the pattern and the density of aluminum. The number obtained for that input mass needs to be divided by 0.9 to account for the losses of the system when melting aluminum.

Before main test runs, the system should be checked for any potential microwave leakage around the chamber, lid and drawer, the waveguide connection to the chamber, the waveguide channel flowing from the SSMG to the chamber connection port and the mullite tube. This sweep of microwave leakage is required to confirm that the system operates within safe parameters.

The sand cast is aligned with the holes of the crucible and chamber through three ridges that restrict movement and allow easy user interaction. The crucible was aligned via visual inspection with the hole at the base of the chamber to ensure flow is strictly through the hole.

In the RECLAIM system, the crucible will align with the hole using a small channel that is created with the crucible design to be inserted into the hole and provide a direct pathway to the sand cast pattern. The RECLAIM system also eliminates the need for a second user due to the mullite tube in the feed channel. This system will also require the user to need only a fire extinguisher and thermally safe gloves to mitigate risks of heat related injuries that can occur dropping the aluminum down the mullite tube. **[Noah 7/21]** Reminder to make distinction between past testing that requires safety procedures while future testing with an automated system is different??

2.2.1.3. Pyrolysis Testing

- Gas leak rate
- Microwave leak rate
- Total contents that are condensed (mass recovery)
- Gas speciation via GC

The pyrolysis system underwent thorough testing to verify COMSOL multiphysics simulations, derived heat exchange, and vacuum integrity.

Vacuum integrity was tested by utilizing the vacuum pump, pulling down to an equlization pressure, and

allowing the system to repressurize. A nominal leak rate was then identified to characterize how much argon would be require to displace any oxygen ingress and maintain the minimum combustion limits. **[charles 7/20]** add data

2.2.2. Inputs, Outputs, and Efficiency

Req: Summary of the inputs, outputs, and efficiency of your recycling process. Please include the RECYCLING SUMMARY TABLE below to summarize these details.

- Explanation of why each material was selected based on the list of materials provided
- Data tables of the amount of input material and its yield after it's been recycled through the system

Recycling Summary Table

Question	**Answer**
Eligible trash item(s) from Table 4 that your solution will recycle.	Aluminum structures/struts, reclosable polyethylene bag
Total mass of one batch of eligible items (expressed in kg). Please explain how the batch will meet Minimum Batch Requirements.	
Mass of each eligible trash item in one batch (expressed in kg). Please explain how the batch will meet Minimum Batch Requirements.	
Length of time (expressed in hours or minutes) to complete one cycle.	
Electricity required for one cycle, including: • Peak demand over a specific time period (kW) • Total electricity consumed (kWh) • Net electricity consumed, if any electricity is produced in the recycling process (kWh)	• Peak: 9kW • Total: **[Noah 7/21]** Need how long each pump runs for and how long SSMG and chiller run for. Use current sensor to track avg power used by control subsystem • Net: 0W
Amount of water (expressed in kg) required for one cycle.	
Types and amounts of other inputs required for one cycle. List each additional input and the mass required for one cycle (expressed in kg).	
Number of team members and time per team member (expressed in minutes or hours) required to operate one cycle.	
Usable outputs produced by one cycle and amounts of each (expressed in kg).	

Question	Answer
Unusable outputs produced by one cycle and amounts of each (expressed in kg).	
Calculate the Mass Efficiency (%) for one cycle. Mass Efficiency is equal to the mass of the usable output(s) divided by the sum of the mass of the trash item inputs and other inputs, excluding energy consumed. Simulated regolith does not count against Mass Efficiency. If your solution uses simulated regolith, you may exclude it from this calculation.	
Calculate the Energy Efficiency (%) for one cycle. Energy Efficiency (kg/kWh) is equal to the mass of the usable output(s) divided by the energy consumed to produce those usable outputs.	**[Noah 7/21]** Need kwh of full cycle

2.2.3. Output Characterization and Verification

Req: Provide a detailed characterization for each of the usable outputs from your recycling process (feedstocks and/or finished products) and a description of how you intend to verify this characterization at the Final Round Demonstration.

- Gas chromatography + flowmeter difference
- Weighing + visual inspection of molded part

The output products are characterized into two domains, metallic derived and plastic derived products.

The metallic derived products mentioned earlier are characterized based on the mass efficiency and product integrity. After a smelting cycle a small portion of metallic material remains within the crucible, this portion was tested below 5-10% loss. Mass input and output are measured to compute mass efficiency of the system and compared to original the original piece for differences in quality. RECLAIM has previously demonstrated screws, bolts, wrenches, nozzles, and ingots. The possibilities are endless, most 3D printable structures could be made into a mold.

For this test we will preheat a crucible at 1kW for five minutes within the chamber. Next, the microwave power will be shut off temporarily for safe addition of the aluminum dropped into the crucible via the mullite tube. The microwave power is then turned on in step increments of 1kW every minute until the system is being fed 3kW of power which varies based on the mass being added to the system. During the competition, the system will be smelting **[this number will be approximated based on the success of Charlies testing]** for X amount of time. Once the required amount of time has passed dependent on the mass, the microwave power is turned off and the vibrator is ran for 5 seconds to allow all molten aluminum to flow into the sand cast pattern. We will demonstrate it works functionally by obtaining a successful cast and comparing the newly created piece to the original.

The pyrolysis derived products are characterized by mass efficiency and product yield. The pyrolysis system will vent all gaseous contents during the final demonstration due to monetary and safety considerations. Before the competition, the gaseous content was characterized using a gas chromatograph setup while utilizing

the expected waste-stream products the team will use in the final demonstration. The contents of this test were filed within the proposal in the pyrolysis test section. The mass quantity will be weighed before testing and measured with a flow meter during testing to determine total output recovery. To determine the total amount of gaseous product produced, the difference between the input of the purge gas is subtracted out of the total flowmeter value. Liquid and solid products will also add to that mass total.



*Figure 15: Successful ilmenite cast using SMELT system **[charles]** take a lil nicer picture of this not in the sand bucket*

2.3. Final Round Schematics

Req: Provide schematics for key elements of your solution, including process design, mechanical and fluids schematics, electrical schematics, and assembly-level drawings showing envelopes and key dimensions. (Recommended length: 4–8 pages)

- P&ID snippets

PLACEHOLDER — BLOCK DIAGRAM **[charles 7/15]** CONOPS / P%ID should be able to cover this "block diagram" so this probably isn't necessary

Figure 16: Block-diagram for process flow

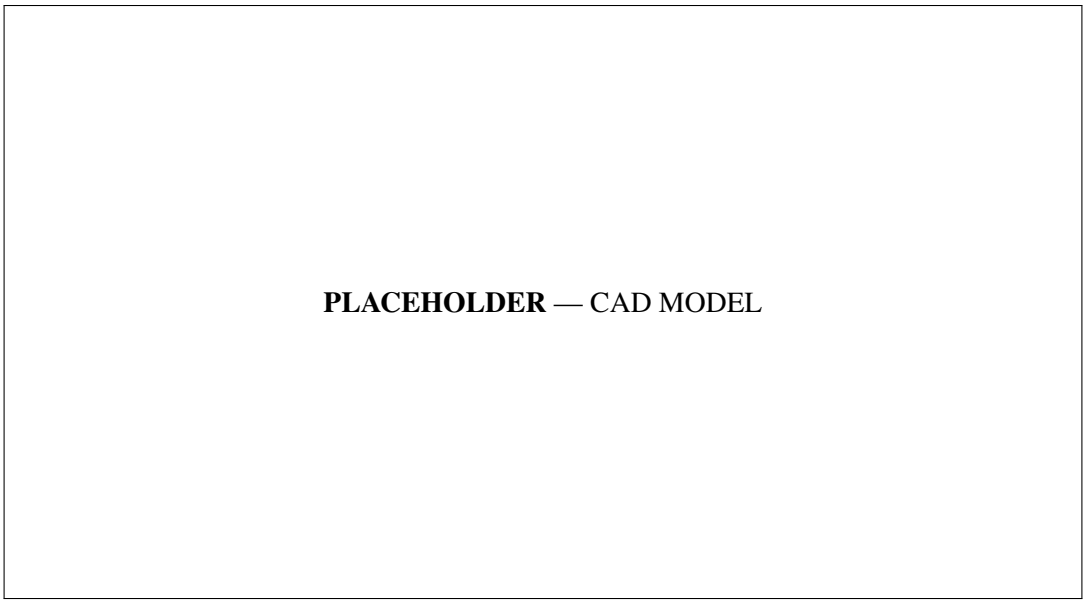


Figure 17: CAD Model of RECLAIM

Description of Equipment and Supplier	**Estimated mass (kg)**	**Estimated volume (cm^3)**	**Maximum allowable working pressure (PSIG)**	**Maximum allowable operating temperature (°C)**
NI cRIO-9024, 800 MHz CPU, 512 MB DRAM, 4 GB Storage CompactRIO Controller (Legacy)	0.609	614.38	Not Listed	55
NI 9263, ±100 kS/s/ch Simultaneous, ±10 V, 4-Channel C Series Voltage Output Module	0.136	142.36	Not Listed	70
NI 9205, ±10 V, 250 kS/s, 16-Bit, 32-Channel C Series Voltage Input Module	0.158	142.36	Not Listed	70
NI 9474, 30 V, 8-Channel (Sourcing Output), 1 μs C Series Digital Module	0.150	142.64	Not Listed	70
NI 9213, 16 channel, up to 75 S/s per channel, ±78 mV C Series Temperature Input Module	0.159	142.36	Not Listed	70
Razzbery Pi 5, 4GB RAM	0.500	168.00	Not Listed	70

Figure 18 serves as the C&DH schematic of the overall system. It features the backbone of the automation architecture for the entire system. These features would include; the systems telemetry data, output devices from the NI 9263 & 9474, and the supporting modules, NI 9213 & 9205 used to help collect and output data for full automation control of the system.

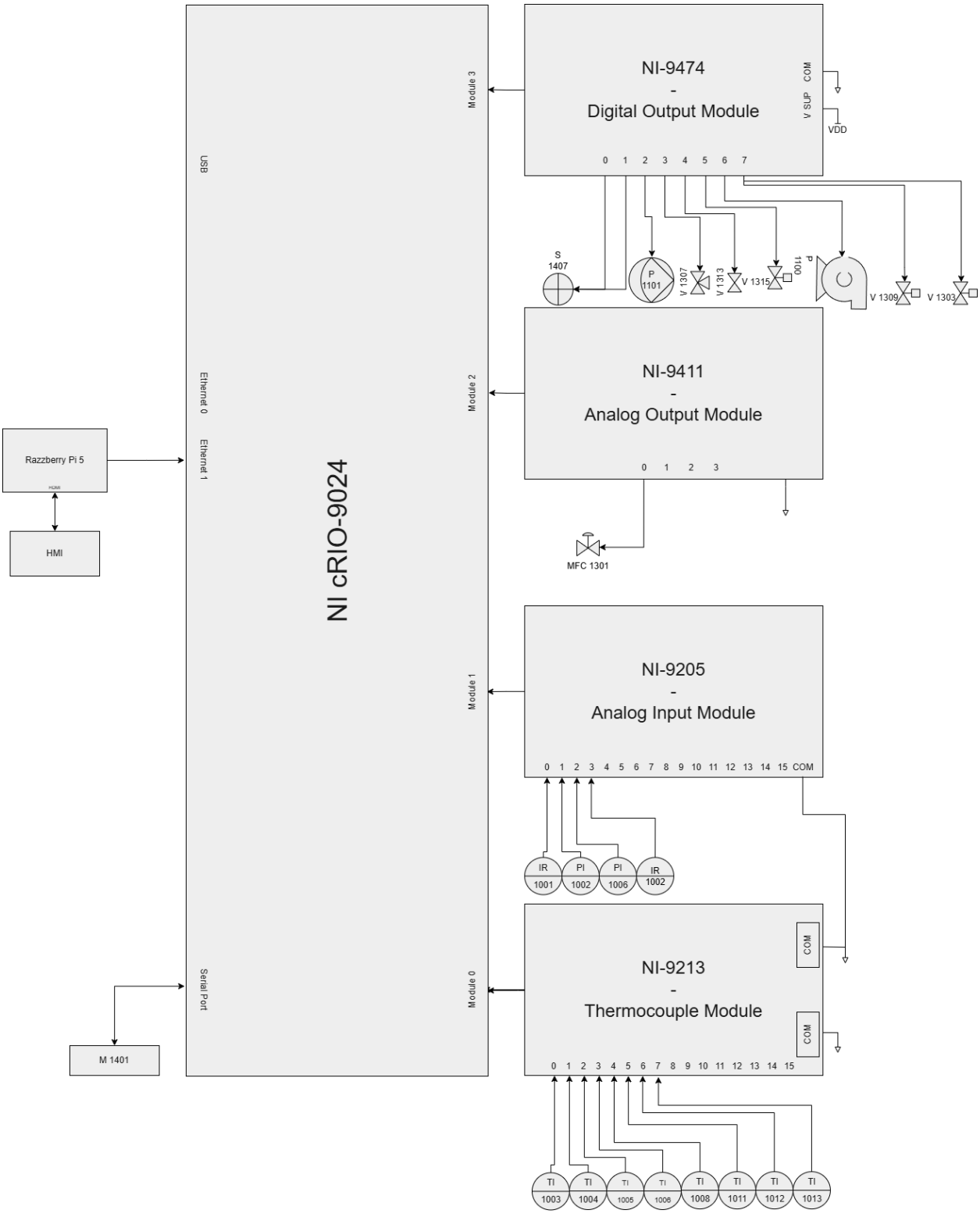


Figure 18: C&DH schematic

Microwave Subsystem Equipment Table

Description of Equipment and Supplier	**P&ID Ref**	**Estimated mass (kg)**	**Estimated volume (cm^3)**	**Maximum allowable working pressure (PSIG)**	**Maximum allowable operating temperature (°C)**
RFHIC Solid State Microwave Generator (RIU256K0-40TG)	M 1401	48.7	74233.152	72.5	50
Daihen Automatic Tuner (SMA-15A)	T 1406	4.7	11,533	N/A	45
Gerling Dummy Load (GA1232)	D 1404	Not Listed	1,100	70	Not Listed
Logus CPR Waveguide Switch	S 1408	Not Listed	1,700	30	Not Listed
Astex Circulator (C13477)	C 1402	Not Listed	2,400	Not Listed	Not Listed
Daihen Waveguide Adapter (WR340 to WR430) x2	A 1403, 1405	Not Listed	Not Listed	Not Listed	Not Listed
Waveguide Adapter (WR430 to WR284)	A 1407	Not Listed	Not Listed	Not Listed	Not Listed
Waveguide Step Transition (WR284 to WR340) x2	A 1409, 1410	Not Listed	Not Listed	Not Listed	Not Listed

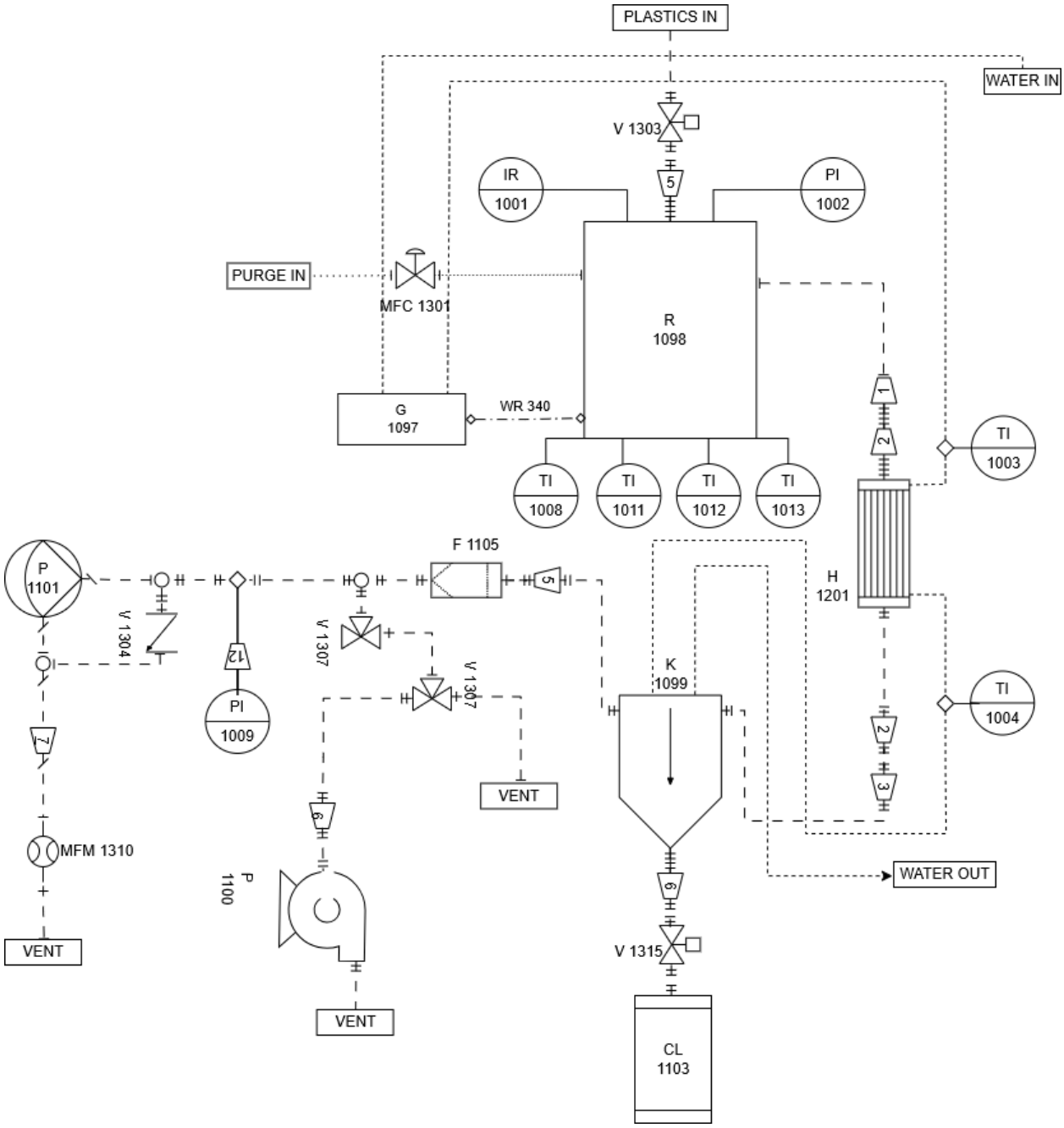
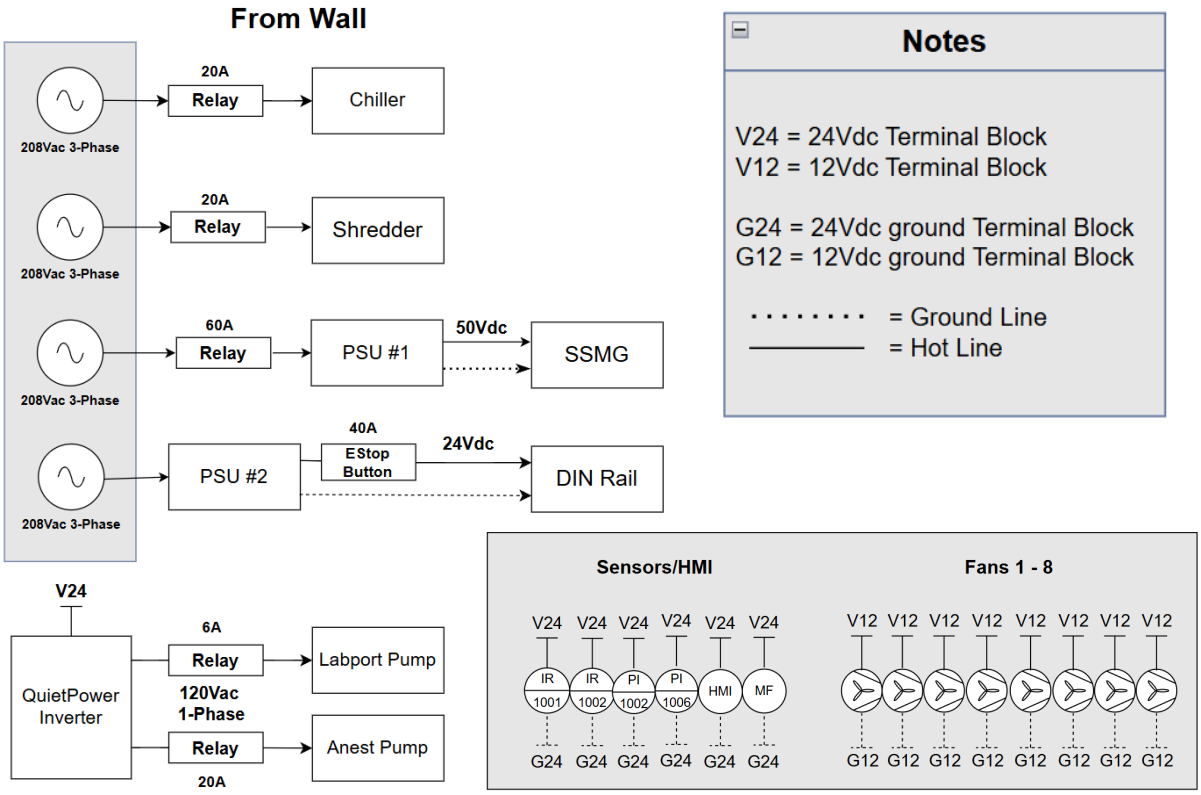
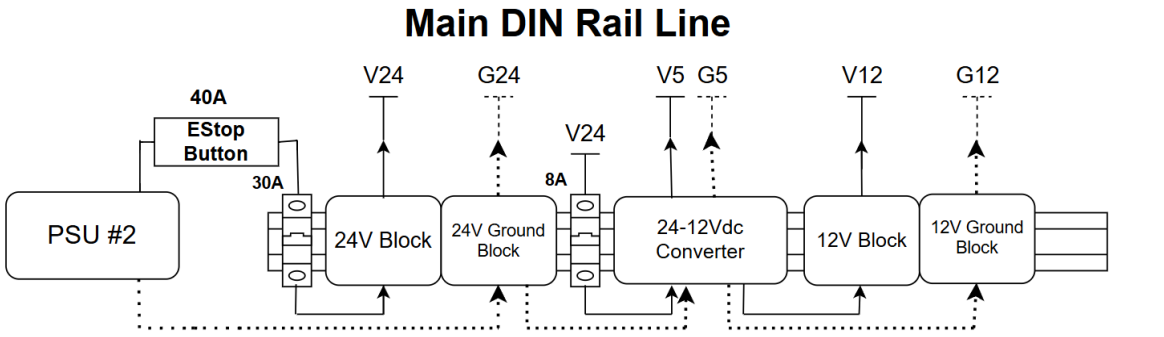


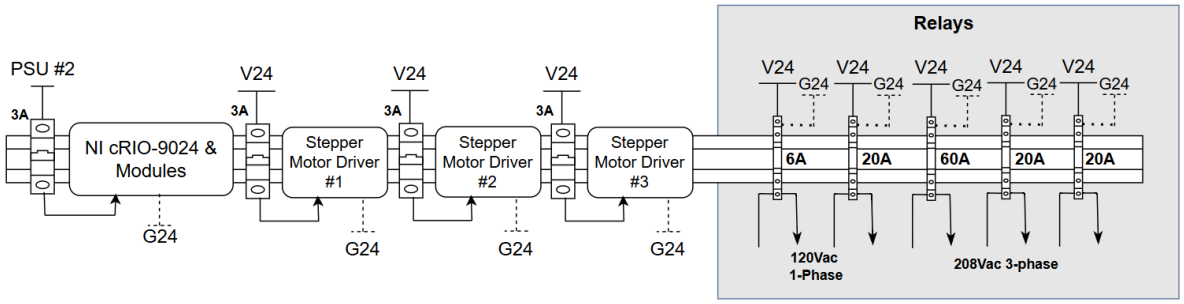
Figure 20: Plastics P&ID ***[charles 7/17]*** *make sure to replace pending final configuration, make sure all labels are cleans and correct, and that all connections are straight, and all flanges / barbs / bolts connections are correct - also add legend - FINISH W/ LUCAS*

Pyrolysis Subsystem Equipment Table

Description of Equipment and Supplier	**P&ID Ref**	**Estimated mass (kg)**	**Estimated volume (cm^3)**	**Maximum allowable working pressure (PSIG)**	**Maximum allowable operating temperature (°C)**
Pyrolysis Reactor (custom, ISO200)	R-1098	TBD	TBD	TBD	TBD
Purge Gas Mass Flow Controller (TBD)	MFC-1301	TBD	TBD	TBD	TBD
Feedstock Gate Valve x2	V-1303	TBD	TBD	TBD	TBD
IR Pyrometer	IR-1001	TBD	TBD	TBD	TBD
Reactor Pressure Transducer (MKS 902B)	PI-1002	TBD	TBD	TBD	TBD
Reactor Thermocouples x4	TI-1008, 1011, 1012, 1013	TBD	TBD	TBD	TBD
Condenser Inlet/Outlet Thermocouples x2	TI-1003, 1004	TBD	TBD	TBD	TBD
Tri-Clamp Baffled Condenser (3"×18")	H-1201	TBD	TBD	TBD	TBD
Knockout Pot / Cold Trap	K-1099	TBD	TBD	TBD	TBD
Product Drain Valve	V-1315	TBD	TBD	TBD	TBD
Liquid Collection Vessel	CL-1103	TBD	TBD	TBD	TBD
Coalescing Oil Filter (5 μm)	F-1105	TBD	TBD	TBD	TBD
Direct/Hose Vent Switching Ball Valve x2	V-1307	TBD	TBD	TBD	TBD
Downstream Pressure Transducer (MKS 902B)	PI-1009	TBD	TBD	TBD	TBD
KNF LABOPORT UN840.3 FTP Pump	P-1101	TBD	TBD	TBD	TBD
Overpressure Check Valve	V-1304	TBD	TBD	TBD	TBD
Purge / Vacuum Simulation Pump	P-1100	TBD	TBD	TBD	TBD
Vent Line Mass Flow Meter	MFM-1310	TBD	TBD	TBD	TBD







Electrical Subsystem Equipment Table

Description of Equipment and Supplier	**P&ID Ref**	**Estimated mass (kg)**	**Estimated volume (cm^3)**	**Maximum allowable working pressure (PSIG)**	**Maximum allowable operating temperature (°C)**
SSMG Power Supply Unit, 50 Vdc (PSU #1)	PSU-1	32kg	36,960,828cm^3	N/A	50 C
DIN Rail Power Supply, 24 Vdc (PSU #2)	PSU-2	33kg	39,828.18cm^3	N/A	50 C
3-Phase Circuit Breaker, 60A (SSMG feed)	—	0.4kg	360cm^3	N/A	70 C
3-Phase Circuit Relay, 20A (Shredder feed)	—	0.375kg	360cm^3	N/A	70 C
3-Phase Circuit Relay, 20A (Chiller feed)	—	0.375kg	360cm^3	N/A	70 C
DIN Rail Circuit Breakers (8A, 5A, 3A x3)	—	0.10kg	90cm^3	N/A	40 C
Stepper Motor Driver	—	.28kg	439.12cm^3	N/A	40 C
Phoenix Contact Relay 6A Single Phase	—	0.04kg	46.43cm^3	N/A	70 C
Sensata Contact relay 20A Single Phase	—	0.28kg	187.97cm^3	N/A	80 C
ESTOP Button 40A Single Phase	—	tbd	tbd	N/A	tbd
IR Pyrometer mini212 and mini22, 24 Vdc	IR-1001	0.39kg	225.8cm^3	N/A	60 C
Pressure Indicator, 24 Vdc	PI-1002 & PI-1006	0.17kg	1.023cm^3	29 PSIG	40 C
HMI Display, 24 Vdc	—	TBD	TBD	N/A	TBD
Cooling Fans, 12 Vdc x8	—	TBD	TBD	N/A	TBD
QuietPower Inverter, 24 Vdc to 120 Vac 1-Phase	—	25kg	65,110cm^3	N/A	60 C
Anest Pump, 208 Vac 3-Phase	—	25.2kg	32,600cm^3	7 PSIG	40 C
KNF LABOPORT Pump (via inverter, 120 Vac)	P-1101	9.6kg	10,892cm^3	15 PSIG	40 C

The system architecture utilizes a centralized 208-Vac 3-phase primary power source, which is branched to support high-load components and a scalable DC control backbone. Primary 208-Vac power is distributed

directly to the SSMG power supply, the water chiller, and the material shredder.

For control and ancillary components, a single-point 208Vac input is fed into a DIN rail-mounted power supply, which converts the input to 24Vdc. This design simplifies system integration, rendering the entire assembly "plug-and-play" for automated environments. The 24Vdc is routed through a circuit breaker to a primary distribution block, powering key instrumentation including the sensors and cRIO chassis. Further downstream, the 24Vdc line is managed through a secondary breaker and a 24Vdc-to-12Vdc converter to support a bank of eight cooling fans. Finally, a dedicated 24Vdc-to-120Vac inverter is integrated within this local distribution circuit to drive two auxiliary pumps. By generating all necessary operating voltages internally, the system minimizes the need for complex site-specific electrical infrastructure, facilitating rapid deployment and seamless integration into automated process flows. **[noah 7/21]** Need to talk about distributing voltage through breakers, relays, and drivers

2.4. Final Round Master Equipment List

Req: Use the following EQUIPMENT TABLE to provide a complete master equipment list, including mass and volume estimates. Includes anything not already referenced in the previous schematics per UA feedback — "should not take up a majority of the report." (Recommended length: 1 page)
[wetz 7/21] Are we including plumbing because its going to waste a lot of space and isn't necessarily critical to our solution as far as specific components (NPT barbed hose fittings, hose clamps, hoses, NPT to NPT adapters, swage to NPT, etc.)

Final Round Master Equipment List

Description of Equipment and Supplier	**Estimated mass (kg)**	**Estimated volume (cm^3)**	**Maximum allowable working pressure (PSIG)**	**Maximum allowable operating temperature (°C)**
Daihen Automatic Tuner Controller (CMC-10)	0.5	600	N/A	45
Julabo Chiller (FL20006)	375	1,758,925	87	40(working), 80(return)
Aluminum Cooling Jacket (Xometry)	5	220	Not Listed	Not Listed
E-Bend Waveguide Turn (WR430)				
Waveguide 90 Twist				
E-Bend Waveguide Turn (WR340)				
E-Bend Waveguide Turn (WR284) x2				
1 1/4 MPT to 3/4 FPT Fittings x2				
3/4 MPT to 3/4 Barbed Hose Fittings x2				
3/4 Barbed Tee Split x2				
3/4 Barb to 3/8 FPT Fittings x4				
3/8 FPT to 3/8 Female Quick-Connect Fitting x6				
3/8 Male Quick-Connect Fitting to 3/8 Barbed Hose Fittings x6				
3/8 Tee Split x2				
3/8 MPT to 1/4 Barbed Hose Fittings x2				
1/4 MPT to 1/4 Barbed Hose Fittings x4				
1/2 MPT to 3/8 FPT Fittings x2				
1/2" Hose				
3/8" Hose				
Hose Clamps x 30				

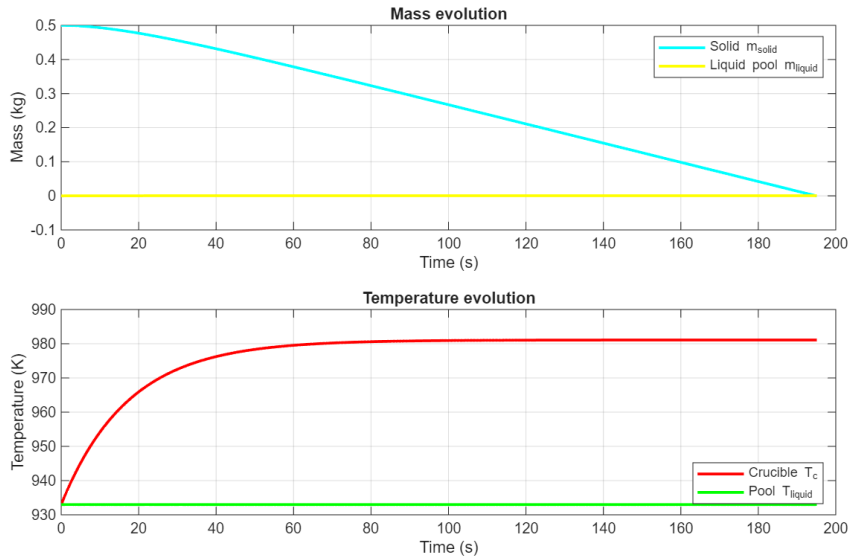


Figure 9: MATLAB model of the flow characteristics after the aluminum has been brought to the melting point and begins to flow through the continuous pour orifice into the sand mold

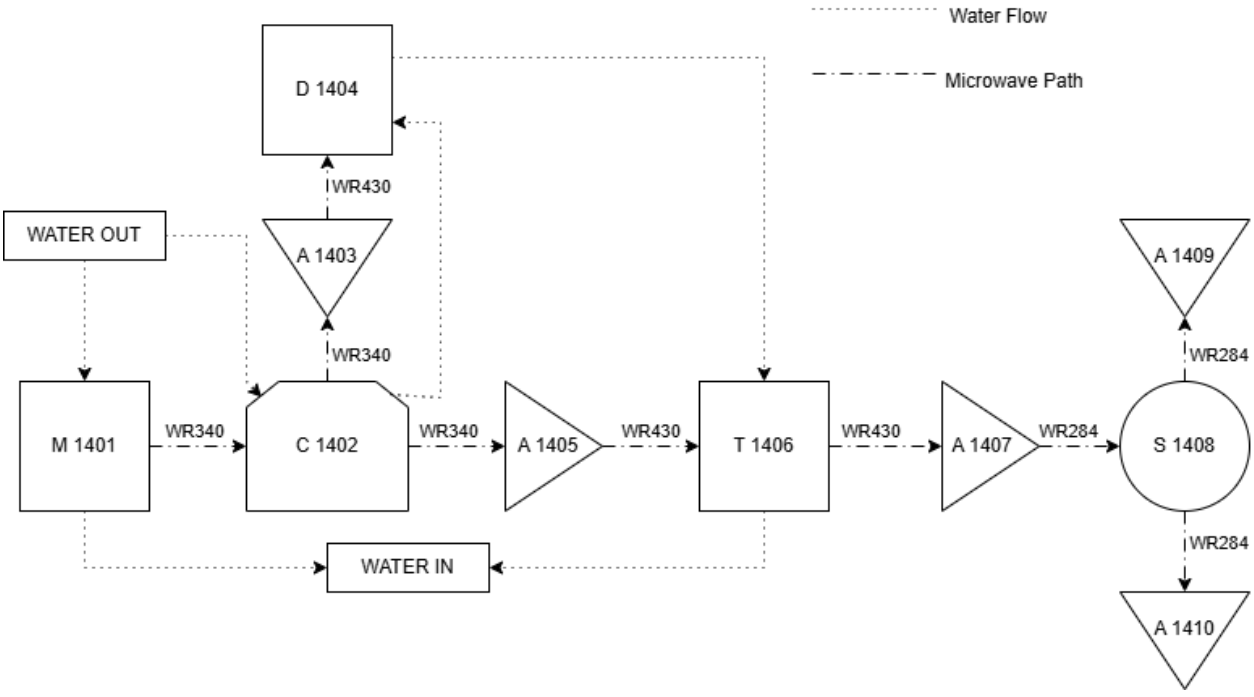


Figure 19: Microwave Subsystem Schematic

3. Pre-Processing of Trash

Recommended length: 1–3 pages

3.1. Demonstrated Pre-Processing

Req: Will you build and demonstrate all hardware related to pre-processing at the Final Round demonstration, or do you intend to use an alternative method at the demonstration site to address pre-processing? If you will build and demonstrate all hardware related to pre-processing, you may skip section 3.2.

- Aluminum strut preprocessing
- **TENTATIVE** Plastics preprocessing

Pre-processing is required for only the aluminum pipe feedstock. Pre-cut aluminum is loaded into the chute of the respective VEVOR shredder, outputting pieces of aluminum roughly 0.5 inches at their longest. The use of the VEVOR shredder allows for versatility between various aluminum inputs.

3.2. Pre-Processing Inputs Table

- **TENTATIVE** if preprocess is integrated into system

Pre-Processing Inputs Table

Parameter	**Description**
Length of time (expressed in hours or minutes) to complete one cycle of any pre-processing included in your design that will not be built and demonstrated	5–10 minutes required for aluminum pipe feedstock
Electricity required for one cycle of any pre-processing included in your design that will not be built and demonstrated, including: • Peak demand over a specific time period (kW) • Total electricity consumed (kWh) • Net electricity consumed, if any electricity is produced in the recycling process (kWh)	Write answer here
Amount of water (expressed in kg) required for one cycle of any pre-processing included in your design that will not be built and demonstrated	None
Amount of other inputs required for one cycle of any pre-processing included in your design that will not be built and demonstrated. List each additional input and the mass required for one cycle (expressed in kg).	None
Number of team members and time per team member (expressed in minutes or hours) required to operate one cycle of any pre-processing included in your design that will not be built and demonstrated.	• One team member will need to be present during the pre-processing of either material • This process will take approximately 5 minutes and (max time needed for plastic shredding)

Appendix

[charles 7/20] do we need an appendix? what is going to go into it?

References

Bai, Y., Li, K., Zhao, J., Yang, C., Bai, Y., Sun, S., & Shang, H. (2026). Microwave-Assisted Catalytic Pyrolysis of Waste Plastics for High-Value Resource Recovery: A Comprehensive Review. *Processes*, *14*(3), 427. https://doi.org/10.3390/pr14030427

Bilén, S. G. (2024). Infrastructure for lunar utilization of microwave energy (ilumen). https://arc.aiaa.org/doi/epdf/10.2514/6.2025-4075

Effinger, M., Wilkerson, R., Voecks, G., et al. (2022). *Scaleup of microwave powder bed fusion for lunar infrastructure construction* (tech. rep. No. NASA NTRS 20220005023). NASA Marshall Space Flight Center. https://ntrs.nasa.gov/citations/20220005023

Ellison, C. R., Elkasabi, Y. M., & Mullen, C. A. (2026). Coupled electromagnetic and thermal analysis of microwave-heated particle beds: Effect of solid volume fraction and particle size. *Thermal Science and Engineering Progress*, *69*, 104463. https://doi.org/10.1016/j.tsep.2025.104463

Gou, M., Chen, Q., Dong, P., Liu, C., & Huang, K. (2023). Design of a microwave heating and permittivity measurement system based on oblique aperture ridge waveguide. *Sensors*, *23*(8), 4035. https://doi.org/10.3390/s23084035

Jiang, Z., Liang, Y., Guo, F., Wang, Y., Li, R., Tang, A., Tu, Y., Zhang, X., Wang, J., Li, S., & Kong, L. (2024). Microwave-assisted pyrolysis—a new way for the sustainable recycling and upgrading of plastic and biomass: A review. *ChemSusChem*, *17*(21), e202400129. https://doi.org/10.1002/cssc.202400129

Liu, R., Lai, S., Hong, T., Zhang, Z., Dong, L., Zhu, H., & Yang, Y. (2024). A low-reflection tuning strategy for three-stub waveguides. *Electronics*, *13*(7), 1304. https://doi.org/10.3390/electronics13071304

McCoy, L. E. (2012). A review: Using pyrolysis and its bioproducts to help close the loop in sustainable life support systems. Retrieved January 1, 2012, from https://ntrs.nasa.gov/citations/20130001565

NASA. (2026). Moonbase phases. Retrieved June 11, 2026, from https://www.nasa.gov/moonbase-phases/

Oliveira, V., Nascimento, D., Castro, L., Souza, T., & Serrão, L. (2021). Aluminum slag separation process analysis through a vibratory machine in the foundry process. *International Journal for Innovation Education and Research*, *9*, 104–120. https://doi.org/10.31686/ijier.vol9.iss5.3075

RFHIC Corporation. (2021). 30 kw, 900 to 930 mhz, solid-state microwave generator [Manufacturer-reported specifications]. https://www.microwavejournal.com/articles/35796-kw-900-to-930-mhz-solid-state-microwave-generator

SMELT Team. (2023). *Smelt: Microwave smelting of metallic feedstocks* (tech. rep.). The Pennsylvania State University. https://bigidea.nianet.org/wp-content/uploads/PSU-Technical-Paper.pdf

Srimurugan, N., & Subbiah, S. (2025). Carbo-thermal reduction of lunar highland regolith simulant for in-situ manufacturing of sic [53rd SME North American Manufacturing Research Conference (NAMRC 53)]. *Manufacturing Letters*, *44*, 416–423. https://doi.org/https://doi.org/10.1016/j.mfglet.2025.06.049

Walter, L., Venkatesh, I., Farbstein, T., Swilley, J., Lowry, E., Callen, T., Withun, P., Krishna, V., Snowdon, K., Chen, A., Stenger, C., Chang, H. S., Lee, J., Walker, J., Kooistra, L., Zahuranic, L., Cano, V., Kim, Y. W., Cardenas, A., … Gollapalli, V. (2025). *Rack and stack: Design of payload racks to support future habitation platforms and exploration missions* (tech. rep.). University of Michigan

- Ann Arbor. https://ntrs.nasa.gov/api/citations/20250005935/downloads/BLiSS_Rack_Stack_FinalReport.pdf

Yamanaka, K., Iyomasa, K., Sugitani, T., Kuwata, E., & Shinjo, S. (2024). GaN solid state power amplifiers for microwave power transfer and microwave heating. *IEICE Transactions on Electronics*, *E107.C*(10), 292–298. https://doi.org/10.1587/transele.2024MMI0004

Zhou, X., Tang, Z., Pedrow, P. D., Sablani, S. S., & Tang, J. (2023). Microwave heating based on solid-state generators: New insights into heating pattern, uniformity, and energy absorption in foods. *Journal of Food Engineering*, *357*, 111650. https://doi.org/10.1016/j.jfoodeng.2023.111650